



Longitudinal Effects of Flywheel Training on Muscle Adaptations and Athletic Task Performance: A Systematic Review and Meta-Analysis with Training Volume Dose-Response Meta-Regression

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ABSTRACT

Background: Training volume is a key variable in resistance training. However, the impact of training volume in flywheel training adaptations has not been thoroughly explored, particularly regarding muscle strength, power, and hypertrophy, as well as performance of athletic tasks such as jumping, sprinting, and change of direction (COD).

Objective: This systematic review aimed to determine the long-term dose-response relationships in flywheel resistance training and provide a robust synthesis of its effects on physical performance.

Methods: Following PRISMA guidelines, four databases were searched for longitudinal studies implementing ≥ 4 weeks of flywheel training and reporting pre- to post-change in overall or eccentric strength, concentric or eccentric power, hypertrophy, jump, sprint, or COD performance. Multilevel meta-analyses were conducted to estimate the overall training effects and investigate the relationship between training volume and the magnitude of adaptations.

Results: Fifty-five studies were included in data synthesis. Large improvements were found in concentric and eccentric power (SMC = 0.943 and SMC = 1.058, $p < 0.01$, respectively). Moderate increases were observed in overall and eccentric strength (SMC = 0.664 and SMC = 0.712, $p < 0.01$, respectively), while hypertrophy, jump, sprint, and COD showed small improvements (SMC = 0.348, SMC = 0.367, SMC = 0.323, and SMC = 0.463, respectively). A dose-response relationship was observed for hypertrophy ($\beta = 0.016$ [0.003, 0.029], $p = 0.018$). Higher inertias improved eccentric power more than lower inertias ($\beta = 50.926$ [1.467, 100.386], $p = 0.044$), though a negative interaction was observed between weekly training volume and inertia ($\beta = -0.727$ [-1.437, -0.017], $p = 0.045$). Bilateral exercises improved overall strength more than unilateral or mixed-limb exercises ($\beta = -0.674$ [-0.949, -0.400], $p < 0.001$; $\beta = -0.571$ [-0.945, -0.198], $p = 0.004$, respectively). Concentric power improvements were greater in highly trained athletes compared to trained athletes ($\beta = -1.858$ [-3.567, -0.148], $p = 0.034$). Flywheel-specific power assessment methods showed a significant difference compared to traditional methods ($\beta = -0.918$ [-1.435, -0.401], $p < .001$). No dose-response relationships were found for overall and eccentric strength, concentric power, jump, sprint, or COD performance ($p > 0.05$).

Conclusions: Flywheel training is particularly effective for improving muscle strength and power. A dose-response relationship was observed for hypertrophy, indicating that higher training volumes enhance adaptations. While higher inertias favor eccentric power, the negative interaction between weekly volume and inertia suggests that higher volumes may attenuate the benefits of higher inertial loads. Training and testing specificity are crucial, as bilateral flywheel exercises produce greater overall strength adaptations than unilateral or mixed-limb configurations, and flywheel-based assessments more accurately capture concentric power adaptations than traditional testing methods. Athletes with higher training levels experienced greater improvements in concentric power than less trained athletes.

Protocol registration: The original protocol was prospectively registered at the Open Science Framework: osf.io/s3qgr.

Key Points

- Flywheel training improves strength and power, with bilateral exercises producing greater strength improvements than unilateral or mixed-limb approaches.
- Concentric power improvements are greater in highly trained athletes and are best assessed using flywheel-specific tests. Higher inertia enhances eccentric power, although high inertia combined with high training volume is counterproductive.
- Flywheel training yields small effects on hypertrophy and athletic performance. Hypertrophy follows a dose–response relationship, whereas improvements in athletic tasks are limited and not volume-dependent, suggesting lower volumes are sufficient.

INTRODUCTION

Effective resistance training programs require careful manipulation of training variables, such as volume, load, rest intervals, exercise selection, frequency, and periodization strategy to elicit the desired adaptations (1). Among these variables, training volume is widely considered one of the most critical factors for maximizing muscle strength, power, and hypertrophy (1-4).

Training volume is commonly expressed as the total sets multiplied by repetitions ($\text{Sets} \times \text{Reps}$), or as total volume load ($\text{Sets} \times \text{Reps} \times \text{Load}$). Regardless of the calculation method, increases in volume generally enhance adaptations in muscle strength, power, and hypertrophy in traditional resistance training (3-5). For instance, Schoenfeld et al. (3) reported that each additional set increased muscle size by 0.37%, while Brigatto et al. (6) found that high-volume group achieved greater improvements in the bench press and squat 1-repetition maximum (1-RM) compared to moderate- and low-volume groups. In contrast, Clemente et al. (7) found no differences between lower and higher volume groups in jumping, sprinting, and change of direction (COD). These conflicting findings highlight the importance of dose-response studies for guiding researchers and practitioners in program design. Furthermore, poor management of training variables may lead to excessive fatigue, delayed recovery, and limited adaptations, highlighting the need to identify more optimal volume dosing strategies.

Flywheel resistance training is an isoinertial training modality that differs from traditional gravity-dependent methods. It utilizes the kinetic energy stored in a rotating flywheel to generate resistance, enabling users to generate greater eccentric forces by delaying maximal braking during the eccentric phase (8). Over the past two decades, research has demonstrated its effectiveness for improving neural and morphological adaptations, including strength, power, hypertrophy, jump, sprint, COD performance, tendon properties, and cardiovascular fitness (9-14). Due to these characteristics, flywheel training also contributes significantly to injury prevention (8).

Despite these benefits, the impact of manipulating training volume on muscle strength, power, hypertrophy, and athletic performance in flywheel training remains unclear. While improvements in 1-RM strength, jump performance, and muscle hypertrophy have been reported (13, 15-19), the findings regarding the impact of training volume are inconsistent. Some studies suggested that lower volumes produce greater strength gains (16), whereas

others favored higher volumes (15). Similarly, sprinting, COD, and hypertrophy outcomes varied across studies employing different volumes (13, 15, 16, 18, 20-23), highlighting the need for a clearer dose-response framework.

These inconsistencies may be due to methodological heterogeneity, including differences in prescribed weekly training volume, inertia (or their potential interactions), participant training levels (recreational or trained), exercise limb allocation strategies (e.g., bilateral, unilateral, or combined exercises), and periodization approaches. Therefore, a meta-analytic approach that controls for these moderators is warranted to clarify the relationship between flywheel training volume and subsequent adaptations.

Although previous reviews have examined the pre–post effects of flywheel training on muscle strength, power, hypertrophy, and athletic performance (24, 25), the number of individual studies has grown substantially in recent years. This expansion often introduces significant heterogeneity and variation in study quality, which leaves the overall certainty of evidence for these outcomes highly uncertain. Furthermore, previous reviews did not specifically investigate the dose-response relationship. Therefore, an updated synthesis is required to integrate recent findings with a rigorous quality assessment (e.g., using GRADE) and to formally explore the dose-response relationships between flywheel training volume and adaptations across all relevant outcomes.

2. METHOD

2.1. Registration of Systematic Review Protocol

A systematic review was carried out per guidelines in the Cochrane Handbook for Systematic Reviews of Interventions (26) and following the 2020 checklist for the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (27). The original protocol was preregistered at the Open Science Framework (osf.io/s3qgr/). Protocol registration occurred after the preliminary, pilot search was completed, but before beginning screening.

2.2. Eligibility Criteria

Studies were included if they met the following eligibility criteria: 1) published in English; 2) published in peer-reviewed journals; 3) participants were free from injury, illness or drug use; 4) participants were between 18 and 40 years old (when specific group ages were not reported separately, the mean age was considered); 5) participants were recreationally active individuals or athletes, 6) had ≥ 4 weeks of flywheel training intervention; 7) studies reported pre–post outcomes in at least one of the following domains: muscle strength, hypertrophy, power, jumping, sprinting, or COD. For muscle strength: isokinetic tests, one-repetition maximum, repetition maximum tests, data collected through strain gauge, linear position transducer, and rotary encoder devices (in Newton [N] and velocity [m/s] units) were considered. For hypertrophy, changes in lean body mass or muscle size (whole muscle or muscle fiber) were considered. For muscle power, changes in watts were considered. For jumping performance, only jump height and reactive strength changes were considered. For sprinting and COD, time and velocity changes were considered; and 8) studies provided sufficient methodological detail for data extraction and reporting of training variables (e.g., studies calculating training volume in seconds, which was not foreseen in the pre-registration, were excluded).

2.3. Information Sources and Search Strategy

A PICO's systematic search strategy was created for the search of databases (Table 1: PICO's of Research). To create and develop search strings, review articles and their references, as well as the Yale MeSH Analyzer (28) were employed. Additionally, to ensure the scope of the search strings, the Word Frequency Analyzer Tool (<https://sr-accelerator.com/#/help/wordfreq>) was utilized. Moreover, a search refiner tool (<https://sr-accelerator.com/#/help/searchRefinery>) was used to determine the sensitivity of the search strings in PubMed, and search strings were adapted to other databases by using the Polyglot Search Translator Tool (<https://sr-accelerator.com/#/help/polyglot>). Therefore, search strings specific to each database were prepared and shared as a supplement document (osf.io/s3qgr/). As the primary search strategy, PubMed/MEDLINE, Scopus, SPORTdiscus, and CINAHL (Cumulative Index to Nursing and Allied Health) databases were used, while the secondary search strategies consisted of 1) screening the references of included studies, 2) review articles on flywheel training, and 3) forward citation tracking of the included studies via Google Scholar. The articles that could not be accessed were searched for via WorldCat (<https://search.worldcat.org/>). Lastly, search alerts were created in all databases prior to the screening process, and alerts were monitored until 02/12/2025. Studies that did not meet the eligibility criteria were excluded.

Table 1. PICO's of Research

Category	Inclusion Criteria
Population	Recreational participants, amateur and elite athletes
Intervention	Flywheel training
Comparator	Post-intervention effects of flywheel training
Outcome	Strength, hypertrophy, power, jump, sprint, change of direction
Study Design	Randomized-controlled trials Non-randomized controlled trials

2.4. Study Selection

The references were uploaded to the reference manager, and duplicate references were removed (EndNote X9.3.3). The articles were screened by two researchers (DB and MB) for eligibility. The reviewers were blinded to reduce bias risk in the screening. The same two authors independently screened titles and abstracts to determine initial eligibility using the systematic review software Rayyan (<https://www.rayyan.ai/>) (29). Thereafter, the full-text screening was independently performed to decide the eligibility for inclusion. All disagreements were resolved with discussion, and the opinion of a third reviewer (IJ) was consulted if required.

2.5. Data Extraction

Standardized Microsoft Word and Excel spreadsheets were prepared to extract and organize information from all included studies. Data on study characteristics including author and year, study design, number of participants,

sex, age, height, body mass, training experience, current training status, and training level, were extracted into a Word document (Table 2). Training-related variables were also extracted into a Word document (Table 3). These included author and year, exercise selection, training volume (sets $\times$ repetitions, including momentum build-up repetitions), intensity prescription method, inertia ($\text{kg}\cdot\text{m}^2$), rest intervals between sets (minutes), periodization strategy, intervention duration (weeks), adherence, and method of assessment. Data required for synthesis were extracted into an Excel spreadsheet as pre- and post-means, standard deviations, raw mean differences, and standard deviation changes. When available, *r* values, *t* values, and exact *p* values were also recorded. To facilitate potential meta-analytic models, all possible quantitative training variables were manually calculated and extracted as continuous moderators. These comprised intervention duration in weeks, average sets per week (i.e., total sets during the training period divided by duration), average volume per week (i.e., total repetitions during the training period divided by duration), average inertia (the sum of all inertial loads divided by the number of different inertial loads used during the training period, when multiple loads were applied). The total volume of the training period was calculated as all repetitions completed during the intervention, while total volume load was calculated using the traditional volume load equation (sets $\times$ repetitions $\times$ inertia). From this, the average volume load per week was derived by dividing the total volume load by duration, and the total volume load for the training period was additionally calculated as all repetitions performed multiplied by the average inertia. When multiple exercises were included, training volume was summed separately for the upper and lower body. In addition to continuous moderators, categorical variables were also recorded, as described in the relevant section below.

If authors reported any missing/insufficient data, the authors (DB and IJ) sent emails to the corresponding authors to complete the information. If needed, a follow-up email was also sent. If sufficient data were not provided, WebPlotDigitizer (Version 4.1, <https://automeris.io/v4/>) was used to extract the raw data from figures where appropriate (performed by DB). Upon completion, ten randomly selected studies were cross-checked by another author (IJ) to identify sources of potential mistakes during data extraction.

2.6. Risk of Bias and GRADE Assessments

Consistent with previous studies, a modified version of the Cochrane Collaboration's tool for risk of bias was used (30, 31). A modifiable assessment tool was chosen to accommodate risk of bias criteria relevant to flywheel training studies. Specifically, in many sport science intervention studies, blinding outcome assessors is impractical, as any form of encouragement or feedback can influence performance. Additionally, blinding personnel and withholding training-related information may pose a safety risk. Accordingly, the assessment tool was modified by removing the criteria "blinding of participants and personnel bias" and "blinding of outcome assessment bias" and adding "effort bias", "familiarization bias", and "training prescription bias" criteria, which are related to the flywheel training studies.

The risk of bias assessment was performed independently by two reviewers (DB and IJ). Any differences in ratings were solved via discussion and subsequently merged into a single document after agreement. The risk of bias assessment interpretation was performed in the frame of the Grading of Recommendations Assessment, Development and Evaluation (GRADE) system. The GRADE approach was applied only to the meta-analytic model without moderators, whereas meta-analytic models with moderators were used as supplementary evidence.

Given the research question—specifically, the use of pre- and post-intervention effects of flywheel training without a control group, which is similar to observational studies—the starting certainty of evidence was rated as low (32). The certainty of evidence was then upgraded to moderate or high, or downgraded to very low, based on several criteria. If the 95% confidence interval of the mean observed effect crossed one category threshold of the Cohen’s *d* effect size (< 0.29), this was considered a violation of precision, resulting in a downgrade of certainty. For inconsistency, if there was high heterogeneity ($I^2 > 75\%$) in the meta-analytic model without moderators, the certainty was downgraded. For risk of bias, if 50% or more of the studies had multiple (≥ 2) high-risk assessments, the certainty was downgraded by one level. For the large magnitude effect, if the studies demonstrated a large effect size (mean observed effect ≥ 0.8) without violating the precision criteria, the certainty was upgraded. Additionally, if a dose-response gradient was observed, the level of evidence was upgraded. In the context of training studies, volume and intensity (with intensity defined as inertia in this manuscript) were considered primary training variables representing dose. Therefore, if the meta-analytic models with moderators showed a significant dose-response relationship ($p < 0.05$) in either volume or inertia, the certainty was upgraded by one level. Other criteria of indirectness, publication bias, and plausible residual confounding were not evaluated in this research due to their low relevance. Indirectness was not evaluated, as only studies directly relevant to the current research’s PICO were included. Thus, all studies provided direct evidence addressing the research question. Publication bias was not assessed, given the tendency of studies not to pre-register their protocols (see Figure 2). As a result, it is not possible to determine whether all relevant findings (both positive and negative) were reported. Plausible residual confounds were also not evaluated. While many training studies report key confounding factors such as age, sex, training status, injury, and health status, and training variables, residual confounders—including diet/nutrition status, motivation/psychological status, supplement and medication use, and sleep quality and quantity—may still influence outcomes. Consequently, the impact of these residual confounders on the results remains uncertain.

2.7. Statistical Analysis

2.7.1. Preparation for Statistical Synthesis

The meta-analysis was structured to quantify pre- and post-intervention effects of flywheel training on outcome variables and their relationship with continuous and categorical moderators. Hence, no comparison group was needed for statistical synthesis. When studies included multiple flywheel resistance groups, data from all groups were extracted for data synthesis, as there was no connection between the intervention groups within the study. However, if a study included both flywheel intervention groups and groups combining flywheel training with other training methods, only the flywheel resistance training group that did not combine flywheel training with other methods was included (see Petré et al. (33)). If a pure flywheel training group was not available, the most relevant flywheel training group was selected for data synthesis (see Sanchez et al. (34)).

2.7.2. Calculation of Effect Sizes, Standard Deviations, and Variance

Standardized mean change with raw score standardization (SMCR) was calculated as the measure of effect, and used for matched pairs outcome measures (35). This approach involved calculating the differences between pre- and post-intervention means, divided by the pre-intervention standard deviation (35, 36) with small sample bias adjustment (*C*).

$$SMC = C \left(\frac{M_{Post} - M_{Pre}}{SD_{Pre}} \right); C_j = C_j = 1 - \frac{3}{4(n_j-1)-1}$$

The standardized mean change magnitude was interpreted as: small effect (0.20-0.49), moderate effect (0.50-0.79), and large effect (≥ 0.80) based on Cohen's classification (37).

If the authors reported standard errors or confidence intervals instead of standard deviations, and the required raw data were not provided upon request, standard deviations were computed using the following equations, respectively:

$$SD = SE \times \sqrt{N}$$

and

$$SD = \sqrt{N} \times \frac{(UL-LL)}{2z}$$

where UL and LL are the upper and lower limits of the CI, N is the sample size, and z is the critical value for the chosen confidence level (e.g., 1.96 for 95%, 1.645 for 90%, 2.576 for 99%).

The pre-post correlation coefficients (r) required to compute the variance were unreported. Consequently, raw data were obtained for 19 studies, 17 directly from the authors (17, 18, 20, 21, 38-50) and two from public online sources (51, 52), and r was calculated using pairwise correlation. For one study in which the author provided data but did not respond to a clarification request (14), the summary statistics reported in the publication were used instead of the shared dataset. First, all outcome measures were classified into sub-categories (i.e., concentric, eccentric, and isometric strength). For outcomes where the r values could not be directly calculated, a median-based imputation was performed for each sub-category using the available data, assuming r values would be more similar within outcome sub-categories. Variance was then estimated using the following equation:

$$Var(SMC) = \frac{2(1-r)}{n} + \frac{(SMC)^2}{2 \cdot n}$$

2.7.3. Statistical Synthesis of Effect Sizes

Most included studies reported multiple effect sizes. To maximize statistical power and avoid data loss from aggregating effect sizes (53), a three-level (multilevel) meta-analysis approach was employed. This approach estimates variance across three levels: sampling error within individual effect sizes (level 1), variance among effect sizes within the same study (level 2), and variance between studies due to differences across studies (level 3). In doing so, it improves the accuracy of variance estimation, overall effect estimates, confidence intervals, and standard errors (54, 55). A multilevel meta-analysis was conducted for all outcome variables separately. For each outcome, two models were specified: one without moderators to estimate pre-post intervention effect sizes, and one with moderators to examine the relationships between effect sizes and model moderators. Statistical significance was set at $p < 0.05$.

2.7.4. Outcome Variables, Continuous and Categorical Moderators

Outcome variables of interest were muscle strength, power, hypertrophy, as well as jump, sprint, and COD performance. For strength and power, we further distinguished specific adaptations: overall strength, eccentric strength, concentric power, and eccentric power. A separate analysis for concentric strength alone was not performed, as overall strength largely reflected concentric adaptation (63 of 69 effect sizes). When multiple

strength outcomes were available, priority was assigned as follows: concentric > eccentric > isometric strength. When multiple concentric strength measures were reported, assessments unrelated to the prime movers of the flywheel exercise were excluded (e.g., concentric knee flexion strength in a study using only flywheel squat exercise, see Coratella et al. (19)). Conversely, if multiple flywheel exercises targeted different prime movers (e.g., squat and hamstring curls), all relevant outcomes were included (see Monajati et al. (56)). This ensured that muscle-specific adaptations were not masked. When multiple assessments of the same prime mover existed, priority was given to 1-repetition maximum (1-RM) > isokinetic measures. For hypertrophy, only one marker per study was extracted. When multiple measures were available, the priority was: magnetic resonance imaging (MRI) > cross-sectional area (CSA) > ultrasound > DXA > other indicators (e.g., pennation angle). For jump performance, separate analyses for squat and drop jumps were not performed due to limited data; instead, jump type was modelled as a categorical moderator. Similarly, sprint course length was treated as a categorical moderator rather than analysed separately. For COD, all available outcomes were included.

Continuous moderators included duration (weeks), sets per week, volume per week (repetitions), total repetitions, volume load per week, total volume load, and average inertia. Early analysis revealed high collinearity among several moderators ($r > 0.7$), which could destabilize coefficient estimates (57). Accordingly, volume per week was retained as the primary volume moderator, and average inertia was included as a second continuous moderator. Their interaction was also tested.

Categorical moderators included training level, limb involvement, and periodization. Training level was categorized as recreational, trained, or highly trained (58). The limb involvement moderator described the exercise limb assignment, categorized as bilateral, unilateral, or a combination of both. Upper versus lower body was considered in preregistration but excluded due to limited upper-body data. Periodization was included as a binary variable (yes/no), given the higher number of periodized flywheel studies.

All moderators were entered into the multilevel meta-analytic model if at least 10 effect sizes per moderator and sufficient data spread were available. When effect size numbers were limited, moderators were prioritized in the following order: volume per week > inertia > training level > limb involvement > periodization. Additional moderators were introduced when possible: (a) power mode (traditional vs. flywheel-specific) for concentric power, (b) jump type (concentric-only squat jump, slow stretch-shortening-cycle (SSC) countermovement jump, fast SSC drop jump) for jump outcomes, and (c) sprint course (<20 m vs. ≥ 20 m) for sprint outcomes. To avoid overweighting, only two sprint lengths were extracted per study: one <20 m (the shortest reported) and one ≥ 20 m (closest to 20 m, preferably 20 m). Non-prespecified moderators were included when they provided insight into adaptation specificity. Finally, a secondary exploratory model was conducted to assess the threshold effects of volume and inertia. Median splits were applied to categorize “higher” vs. “lower” volume and inertia, and estimates were compared across these thresholds. This analysis was conducted to determine whether a dose-response trend could be observed within the dataset.

2.7.5. Model Selection

Multilevel models with different moderators were first fit using maximum likelihood (ML) estimation. Likelihood ratio (ANOVA) tests were first applied; if not statistically significant ($p > 0.05$), model selection was based on corrected Akaike Information Criterion (AICc; due to small sample size, $n/k < 40$) (59), Bayesian Information Criterion (BIC), and residual heterogeneity (QE). Moderators were only removed if their exclusion improved model fit (both $\Delta\text{AICc} > 10$ and $\Delta\text{BIC} > 10$). Otherwise, the full model was retained, as it was theoretically more relevant and explained more heterogeneity (QE). Final models were then refitted using restricted maximum likelihood (REML) for more accurate variance estimation.

2.7.6. Influential Diagnostics and Sensitivity Analysis

Upon fitting final models, influence diagnostics were performed. Studentized residuals, hat values, and Cook's distances were calculated. Studies exceeding ± 3 for studentized residuals or three times the mean for hat or Cook's distance were flagged. Sensitivity analyses compared models with and without flagged studies, and differences were reported accordingly.

2.7.7. Statistical Heterogeneity

Heterogeneity was quantified using τ^2 (absolute heterogeneity) and I^2 (relative heterogeneity). In the multilevel models, residual heterogeneity was partitioned into within-study and between-study components. For models including moderators, pseudo- R^2 values were calculated to indicate the proportional reduction in heterogeneity (within- and between-study τ^2) attributable to the moderators.

2.7.8. Statistical Analysis Software and Data Availability

All meta-analytical calculations were performed in the *metafor* package in R language and environment for statistical computing (v 4.0.2; R Core Team, <https://www.r-project.org/>). All scripts and Excel extracted data spreadsheets were shared on the Open Science Framework (osf.io/s3qgr/).

3. Results

3.1. Search Results

The primary search yielded 1,582 studies, of which 34 were included. Secondary search strategies (i.e., forward citation tracking and review articles) identified 4,229 studies, with seven eventually included. An additional 388 studies were identified through database alerts, resulting in 17 included studies. Therefore, a total of 58 studies were included in this systematic review. Figure 1 illustrates each stage of the screening process.

PRISMA 2020 flow diagram for new systematic reviews which included searches of databases, registers and other sources

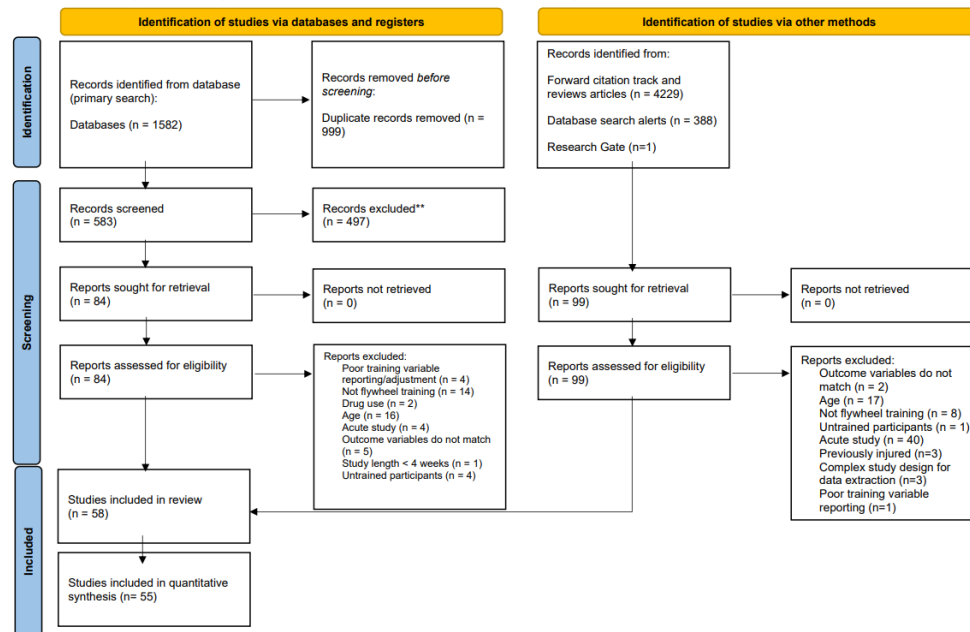


Figure 1. The flow chart of the literature search. n = number of studies.

3.2. Study Characteristics and Main Training Variables

Three studies were not included in the data synthesis for the following reasons: 1) complex study design made data extraction more difficult (60), 2) only the percentage change analysis was performed with no extractable data (61); and 3) pre-intervention means and standard deviation values were not reported (62). Therefore, a total of 58 studies were included in the review, with 55 studies and 716 participants incorporated into the data synthesis. The most frequently used exercise was the squat and its variations (including the half-squat), reported in 39 studies. Single-leg squat and lunge variations were the second most common, appearing in 16 studies. The Romanian deadlift and leg curl exercises were used in eight studies, while knee extension was reported in four studies. Front and lateral step exercises were used in three studies. Hip extension and leg press exercises were each used in two studies. Supine squat, hip thrust, hamstring kicks, shoulder lateral raises, internal and external rotation, upper body press, upper body pull, upper body push press, elbow flexion, and elbow extension exercises were each reported in only one study. The preferred set range was two to six, with repetitions ranging from 4 to 15. Inertial loads varied between 0.025 kg·m² to 0.31 kg·m². Most studies (40) adjusted intensity using a predetermined (pre-allocated) protocol. Concentric or eccentric-based optimal power loading intensity prescription methods were used in eight studies, while velocity- and rate of perceived exertion-based methods were each used in two studies. Study durations ranged from 4 to 39 weeks. More detailed descriptions of participants and study characteristics are presented in Table 1 and Table 2.

Table 2. Study Characteristics

Study (year)	Study Design	Participants	Sex: (M/F)	Age (years)	Height (cm)	Body Mass (kg)	Training Experience	Training Status during or before intervention	Level
Dilek et al. (2025) (63)	Randomized trial	Traditional set flywheel group: n = 5 Rest-redistribution flywheel group: n = 6	0/11	22.1 ± 2.7 22.6 ± 5.3	170 ± 5.9 169.5 ± 7.1	65.5 ± 10 63.6 ± 8.1	Unclear	Three tactical handball sessions and two isoinertial strength sessions	University handball players
Peng et al. 2025 (64)	Randomized parallel- controlled trial	Flywheel group: n = 9	0/9	20.7 ± 2	167.9 ± 4.7	61.9±4.7	≥ 3 years of basketball training and ≥ 2 years of resistance training experience	Three technical and tactical basketball sessions and two physical conditioning sessions	University basketball players
Ćorilić et al. 2025 (65)	Randomly divided	Flywheel group: n = 10	10/0	24.9 ± 5.59	182.9 ± 4.25	73.4 ± 10.23	≥ 3 years of tennis training experience	Regularly train and attend tennis training sessions four times/week	Active tennis players
Chen et al. 2025 (66)	Matched, parallel trial, randomly assigned	Horizontal vector flywheel group: n = 8 Vertical vector flywheel group: n = 8	14/2	20.3 ± 2.5 20.7 ± 2.2	174.1 ± 8.6 178.6 ± 11.4	65.8 ± 7.8 71.9 ± 12	12 ± 4.1 8.4 ± 3.8	Swimming sessions, dry-land training, and competitions	National-level competitive swimmers
Alcántara et al. 2025 (67)	Quantitative quasi- experimental design, randomly divided	Flywheel group: n = 5	0/5	20.33 ± 1.86	167 ± 4	71.83 ± 10.09	6.5 ± 2.17 years of soccer training experience	Three soccer- specific training sessions lasting one hour and 30 minutes/week, and one league game/week	Semi-professional soccer players
Yuan et al. 2025 (68)	Random allocation	Flywheel group: n = 8	8/0	21.57 ± 2.23	179.86 ± 6.04	75 ± 7.05	11.43 ± 1.81	Five tactical and technical sessions/week lasting two hours and match simulations	Elite badminton athletes
Moreira et al. 2025 (69)	Randomly assigned	Flywheel group: n = 10	10/0	19.8 ± 1.22	Unclear	Unclear	Unclear	Moderate physical activity/three days	Recreational participants

								per week on average	
Shimizu et al. 2025 (38)	Randomized controlled trial	Flywheel group: n = 11	11/0	19.5 ± 1.2	178.2 ± 6.5	73.2 ± 7.6	≥ 6 years of basketball experience	Strength training participation once/week, five basketball training sessions/week lasting two hours each, and one competition/week	Collegiate basketball athletes
Spudić et al. 2024 (40)	Randomized parallel-group trial	Flywheel group: n = 18	7/11	23.8 ± 2.6	172.9 ± 9.7	67.6 ± 11.9	≥ 1 year of consistent resistance training experience	Strength training participation at least twice a week	Physically active adults
Xie et al. 2024 (70)	Randomly assigned	Flywheel group: n = 9	0/9	20.7 ± 2	167.9 ± 4.7	61.9 ± 4.7	≥ 3 years of basketball and ≥ 2 years of resistance training experience	Basketball and resistance training participation	College basketball athletes
Hu et al. 2024 (15)	Randomized two-group design with repeated measures	Flywheel group: n = 7	7/0	23.2 ± 0.6	181.5 ± 3.7	77.4 ± 3.6	5.1 ± 0.9	Two sessions of speed, one session of physical fitness exercises and aerobic recovery, and two sessions of strength training	Elite academy sprinters
O'Brien et al. 2024 (39)	Randomly assigned	Flywheel group: n = 10	10/0	18.4 ± 0.6	179.4 ± 4.1	76.6 ± 5.7	≥ 12 months of traditional resistance training experience	Two sessions/week resistance training, three sessions/week soccer training	U-20 soccer players
Asencio et al. 2024 (20)	Unclear	Constant inertia group: n = 9 Variable inertia group: n = 8	17/0	22.9 ± 7.2 22 ± 5.71	180 ± 4 182 ± 8	75.66 ± 6.13 76.20 ± 6.40	≥ 2 years of resistance training experience	Soccer technical and tactical, and strength training sessions	Spanish third division amateur football players
Wang et al. 2024 (51)	Randomized two-group design with repeated measures	Flywheel complex training group: n = 7	0/7	21 ± 2.3	179.8 ± 5.9	67.4 ± 8.8	Unclear	≥ four sessions of volleyball training with resistance training and jump exercises/week	Division I volleyball athletes

Low et al. 2023 (71)	Randomly assigned	Flywheel group: n = 11	Unclear	20.75 ± 1.14	174.23 ± 5.14	168.73 ± 6.79	≥ 4 years of badminton experience	Two to four hours of badminton training/week in the last two years	University badminton athletes
de Keijzer et al. 2023 (72)	Randomized-controlled trial (within-subject, between limbs)	Flywheel group: n = 11	11/0	22 ± 2	174 ± 9	77.2 ± 11.3	≥ 6 months of resistance training experience	≥ two sessions of resistance training/week	Amateur university athletes
Saez de Villareal et al. 2023 (73)	Randomly assigned	Flywheel group: n = 12	12/0	20.5 ± 2.5	179.7 ± 3.7	81.2 ± 5.2	6.5 ± 2.8	Five days of handball training sessions/week	National second division handball players
Murton et al. 2023 (41)	Randomized-controlled trial	Flywheel group: n = 8	8/0	18 ± 1 ^a	Unclear	93 ± 13.1 ^a	≥ 1 year of resistance training experience	Two resistance training sessions/week, three technical sessions/week, and one match/week	Elite club academy athletes
Petre et al. 2023 (33)	Randomized-controlled trial	Flywheel group: n = 7	7/0	24.7 ± 4.3 ^a	181.9 ± 5.6 ^a	81 ± 9 ^a	≥ 2 years of resistance training and sport-specific training on ice	Total of eight to ten training sessions/week	Elite bandy players
Xu et al. 2023 (42)	Prospective cohort trial with a control period	Flywheel group: n = 13	8/5	Men: 19.63 ± 1.6 Women: 19.4 ± 1.52	176 ± 2.2 173.6 ± 4.1	78 ± 13 80.3 ± 13.4	≥ 1 season of collegiate water polo experience	Water polo specific training sessions two to three times/week and one competitive game/week	Collegiate club water polo team
Jarosz et al. 2023 (12)	Randomly assigned	Forward split-squat group: n = 8 Lateral split-squat group: n = 8	16/0	19 ± 1 20 ± 1	177 ± 6 176 ± 10	66 ± 5 70 ± 11	6 ± 1 years of soccer training experience ≥ 2 years of resistance training experience	Field training sessions five days/week, including the match	Amateur soccer players
Puustinen et al. 2023 (74)	Non-randomized parallel group design	Flywheel group: n = 9	9/0	18.9 ± 1	180 ± 7.8	80 ± 9.1	Several years of resistance training experience	Ice hockey and endurance-type training	Elite Finnish hockey players
Filetti et al. 2022 (44)	Randomized-controlled design	Flywheel group: n = 7	7/0	18 ± 1 ^a	182 ± 6 ^a	76.2 ± 6.9 ^a	≥ 7 years of professional football experience	Soccer training sessions four times/week, total soccer training	Professional U-19 soccer players

								volume eight hours/week, 50 minutes of resistance training once/week, and one competitive game	
Maroto-Izquierdo et al. 2022 (61)	Randomly assigned	Flywheel group: n = 10	10/0	21.4 ± 2.2	175.8 ± 5.9	75.1 ± 8.9	No regular lower limb strength training experience	Recreational physical activity six to eight hours/week	Undergraduate sport science students
Pecci et al. 2022 (21)	Randomized-controlled trial	Flywheel group: n = 10	0/10	20.8 ± 2.6	161 ± 4	57.5 ± 7.3	8.86 ± 3.36 years of competition experience ^a	Three soccer field training sessions lasting 90-120 minutes/week, and a competitive match	Professional first division soccer players
Maroto-Izquierdo et al. 2022 (75)	Randomly assigned	Flywheel group: n = 9	9/0	24.9 ± 4.5	191.3 ± 4.3	95.9 ± 11.1	Extensive experience with free-weight resistance training	Four handball sessions (nine hours)/week, three resistance training sessions/week, and one competitive game	Professional handball players
Weng et al. 2022 (76)	Parallel randomized controlled trial	Flywheel group: n = 12	12/0	22.2 ± 2.6	177.6 ± 1.2	71.3 ± 7.4	5.2 ± 1.6	Unclear	Young long-distance runners
Morencos et al. 2022 (45)	Randomized-controlled trial (within-subject)	Flywheel group: n = 17	17/0	22.8 ± 2.5	175.6 ± 5.4	75.2 ± 7.9	≥ 3 years of strength training experience	≥ two resistance training sessions/week	Sport science students
Asencio et al. 2022 (43)	Stratified, randomly assigned	Vertical flywheel group: n = 11 Horizontal flywheel group: n = 8 Mix flywheel group: n = 10	29/0	22.6 ± 4.0 22 ± 4.1 24.9 ± 6.3	179 ± 9 176 ± 5 177 ± 9	75.5 ± 14.1 74.9 ± 11.4 75.6 ± 16.8	≥ 12 years of experience in their respective sports	Three sessions of on-court sport-specific training/week	Recreational athletes

Fousekis et al. 2021 (77)	Randomized three-group repeated measures design	Flywheel group: n = 11	Unclear	24 ± 6.6	180 ± 4.2	77 ± 4.4	2 years of free-weight training experience and ≥ 5 years of training experience	Team soccer training sessions four times/week, and one football match	Amateur soccer players
Martin et al. 2021 (46)	Randomly assigned	Flywheel group: n = 12	12/0	18.8 ± 0.7	181.4 ± 3.5	84.1 ± 2.4	≥ 3 years of active training and strength training experience	Five technical and tactical water polo sessions/week, five sessions of physical conditioning, including two sessions of strength training and three sessions of swimming training/week, and one competitive play/week. Total training volume of twelve hours/week.	Second and top division water polo athletes
Arsenis et al. 2021 (78)	Randomly divided	Flywheel group: n = 16	Unclear	21.31 ± 1.25	179.06 ± 6	75.81 ± 7.35	≥ 6 years of soccer training experience	Soccer training sessions four to five times/week, and strength training participation (1- RM- 1.5 times $\times$ body weight)	Well-trained amateur soccer players
Timmins et al. 2021 (60)	Randomized interventional cohort trial	Flywheel group: n = 13	13/0	22 ± 3	184 ± 7	78.3 ± 6.8	>2 years of experience in hamstring resistance training in a high-level sporting context	Required team, strength, and other training sessions	Semi-professional and fully professional Australian football players
Monajati et al. 2021 (56)	Two parallel group randomized controlled pre-post design	Flywheel group: n = 10	5/5	22.6 ± 2.33	175.3 ± 7.38	69.9 ± 8.26	Unclear	Two volleyball training sessions/week	Recreationally trained volleyball players

Arsenis et al. 2020 (16)	Randomized, three-group repeated measures design	Flywheel group: n = 16	16/0	20.63 ± 1.45	178 ± 4.14	75.94 ± 7.80	Lower body training experience	Three sessions of resistance training/week (1-RM- 1.5 times × body weight)	Well-trained athletes
O'Brien et al. 2020 (17)	Randomized trial	Flywheel group: n = 11	0/11	23.17 ± 5.55	171.41 ± 6.37	77.93 ± 19.05	≥ 2 years of resistance training experience	Three sessions of basketball training/week and two sessions of strength/power training sessions/week	Amateur basketball players
Presland et al. 2020 (79)	Randomly assigned	Conventional flywheel group: n = 10	10/0	26.4 ± 4.1	179.6 ± 6.4	81.5 ± 13.5	Recreational resistance training experience	Unclear	Recreationally active participants
Cabanillas et al. 2020 (62)	Randomly assigned	Flywheel group: n = 4	Unclear	21.3 ± 3.45 ^a	193.5 ± 8.04 ^a	87.43 ± 11.65 ^a	2 years of resistance training experience	Weekly competition participation	Professional basketball players
Sagelv et al. 2020 (52)	Randomized-controlled trial	Flywheel group: n = 13	13/0	23.07 ± 3.15	181 ± 5.0	78.6 ± 7.42	Unclear	Two to three football training sessions/week, one friendly football match/week, and 2.3 ± 1.5 hours resistance training volume/week	Reactionally football players
Coratella et al. 2019 (19)	Pre-post, parallel two-group, randomized controlled trial	Flywheel group: n = 20	20/0	23 ± 4 ^a	180 ± 0.11 ^a	77 ± 5 ^a	≥ 5 consecutive years in youth or semi-professional soccer team experience	Four football training sessions lasting two hours each/week, and one match/week	Italian fourth division professional soccer players
Sabido et al. 2019 (23)	Stratified, randomly assigned	High inertia group: n = 7 Low inertia group: n = 8	4/3 5/3	25.7 ± 10 29.8 ± 12.8	174 ± 11 173 ± 12	80.6 ± 9.4 84.7 ± 16.3	Unclear	Two hours of training sessions/week and one competitive match	Amateur rugby players
Sanchez et al. 2019 (34)	Randomized-controlled study design	Flywheel + HIIT Group: n = 12	12/0	22.5 ± 2.2 ^a	175.8 ± 7.1 ^a	72.6 ± 9.1 ^a	Unclear	Technical/tactical training four to six hours/week	Amateur team sport players

								and one competitive match/week	
Maroto-Izquierdo et al. 2019 (80)	Randomly divided	Traditional flywheel Group: n = 10	Unclear	21.7 ± 3.4^a	177 ± 5.4^a	75.8 ± 9.8^a	No history of regular lower limb strength training	Recreational physical activity six to eight hours/week	Undergraduate sport science students
Hernandez Davo et al. 2018 (81)	Unclear	Bilateral flywheel squat Group: n = 5 Unilateral flywheel squat Group: n = 5	Unclear	21.6 ± 2.4^a	180 ± 9^a	83.6 ± 21.4^a	Unclear	Three sessions of technical and tactical sessions lasting 90 minutes/week	Amateur basketball players
Nunez et al. 2018 (14)	Unclear	Lateral lunge flywheel group: n = 14 Squat flywheel group: n = 13	27/0	22.8 ± 2.9 22.6 ± 2.7	177.3 ± 3.7 164.2 ± 7	75.3 ± 8.8 79.5 ± 12.8	Unclear	Three to four sessions training	Team sports players or sport science students
Illera-Domínguez et al. 2018 (13)	Quasi-experimental design with repeated measures	Flywheel Group: n = 10	8/2	23 ± 4.1	174.8 ± 7.7	71 ± 6.8	Unclear	Self-reported weekly moderate intensity activity 6.3 ± 3.4 hours/week	Physical education students
Skok et al. 2017 (82)	Controlled and randomized study design	Squat flywheel group: n = 19 Multidirectional flywheel group: n = 19	38/0	20.2 ± 1.1 20.8 ± 2.6	179.7 ± 6.5 181.7 ± 5.5	73.4 ± 11.2 75.2 ± 7.6	1 to 3 years of resistance training experience	six hours of training/week and competition participation/week	Semi-professional and amateur team-sport players
Sabido et al. 2017 (83)	Unclear	Flywheel group: n = 11	Unclear	23.9 ± 3.8^a	183 ± 7^a	79.5 ± 7.7^a	10.3 ± 3.4 years of handball experience	Four training sessions/week	First national handball division players
Maroto-Izquierdo et al. 2017 (22)	Randomized-controlled study	Flywheel Group: n = 15	15/0	19.8 ± 3.87	186 ± 8	82.3 ± 12.78	Highly experienced in free-weight resistance training	Four Handball training sessions (nine hours)/week, three strength/power training sessions/week, three on-track specific physical training	Professional first division handball players

								sessions/week, and one competitive match/week	
Sanz-Lopez et al. 2017 (9)	Controlled trial	Flywheel group: n = 9, included limbs: n = 18	9/0, 18/0	22.8 ± 4.2 ^a	179.8 ± 7.9 ^a	72.57 ± 6.73 ^a	Unclear	> three days of sports activity/week	Healthy, active participants
Gual et al. 2016 (47)	Randomized-controlled trial	Flywheel Group: n = 27	16/11	22.5 ± 3.8	183.9 ± 8.8	78.2 ± 8.4	Unclear	3.5 sports training sessions/week	Spanish first and third division basketball and volleyball players
Sanz-Lopez et al. 2016 (10)	Controlled trial	Flywheel Group: n = 9	9/0	22.8 ± 4.2 ^a	179.8 ± 7.9 ^a	72.6 ± 6.7 ^a	Unclear	> three days of sports activity/week	Healthy, active university students
De Hoyo et al. 2015 (84)	Randomly allocated	Flywheel Group: n = 11	11/0	22 ± 2	176.63 ± 2.27	77.47 ± 8.02	Unclear	> thirty minutes and > five times moderate physical activity/week	Healthy, active participants
Lundberg et al. 2014 (18)	Unclear	Flywheel Group: n = 10	10/0	26 ± 5	183 ± 7	77 ± 9	Some of them had resistance training experience	Recreational exercise participation three days/week	Moderately trained college students
Fernandez- Gonzalo et al. 2014 (48)	Unclear	Flywheel Group: n = 32	16/16	Men: 23 ± 4 Women: 24 ± 4	178 ± 8 164 ± 4	75 ± 8 59 ± 8	Unclear	Recreational physical activity four to six hours/week	Sport science students
Lundberg et al. 2013 (49)	Unclear	Flywheel Group: n = 10	10/0	25 ± 4	184 ± 6	83 ± 3	Moderate resistance training experience	Moderately trained, engaging in recreational physical activities two days/week	Moderately trained college students
Seynnes et al. 2007 (50)	Unclear	Flywheel Group: n = 7	5/2	20 ± 0.7	179.3 ± 7.49	74.6 ± 8.07	Unclear	Unclear	Recreationally active participants
C. Askling et al. 2003 (85)	Randomly assigned	Flywheel Group: n = 15	15/0	24 ± 2.6	182 ± 0.06	78 ± 5	Unclear	Unclear	Elite Athletes (Swedish Premier Division Teams)

^aaverage values, n number of participants, m male, f female, 1-RM one-repetition maximum, HIIT high-intensity interval training.

Table 3. Summary of Main Training Variables

Study (year)	Exercise Selection	Volume (Set × repetition ranges)	Inertia (kg·m ²)- (ranges and averages)	Inertia Prescription Method	Sessions	Rest Intervals Between Sets (minutes)	Periodization Strategy	Duration	Adhesion	Outcome Variables (method of assessment)
Dilek et al. 2025 (63)	Standing and prone leg curl	3 × 3-9	0.029	Pre-determined	2 sessions	1 minute and 15 seconds	Constant volume and intensity	8 weeks	Unclear	Concentric and eccentric knee flexor strength at 60/180/300°/s ⁻¹ (isokinetic dynamometer); Countermovement and squat jump (mobile application); 20 m sprint test; Illinois agility test (photocells)
Peng et al. 2025 (64)	Squat	4 × 10	0.075	Pre-determined	2 sessions	3 minutes	Constant volume and intensity	6 weeks	Unclear	20 m sprint test time (timing systems); 5-0-5 change of direction and lane agility test time (timing system)
Ćorilić et al. 2025 (65)	Half squat, Romanian deadlift, and Bulgarian split squat	4 × 6-8	Unclear	Unclear	2 sessions	Unclear	Unclear	6 weeks	Unclear	Squat jump height, countermovement jump height (wall touch scaled in centimeters); 5-meter, 20-meter clay court sprint time (light gates); T-test and 5-0-5 change of direction test time (light gates)
Chen et al. 2025 (66)	Horizontal vector flywheel group: hip thrust Vertical vector flywheel group: half squat	4 × 4-12	0.035-0.085 0.07 (average)	Pre-determined	3 sessions	2 minutes	Volume decremental and intensity incremental	8 weeks	Unclear	15 and 25 m swim sprint test velocity (video analysis); Countermovement and squat jump height (force plates)
Alcántara et al. 2025 (67)	Front step and crossed lateral step acceleration	2 × 6	0.026	Pre-determined	2 sessions	2 minutes	Constant volume and intensity	8 weeks	90%	10 and 20 m sprint time (wireless photocell gates); 8.5 and 17-m curved sprint time (wireless photocell gates); V-cut change of direction test time (wireless photocell gates)
Yuan et al. 2025 (68)	Squat, deadlift, and lunge	4 × 8	Pro, large, medium, small plates	Set RPE-based dynamic intensity adjustment	2 sessions	3 minutes	Constant volume and individualized intensity	6 weeks	100%	5-0-5 change of direction test time (timing gates); Isometric mid-thigh pull test (force plate)

			Set RPE method load allocation (8/10)							
Moreira et al. 2025 (69)	Knee extension	4 × 8-12	0.055	Pre-determined	3 sessions	2 minutes	Volume incremental and constant intensity	8 weeks	Unclear	Maximum voluntary isometric knee extension strength (load cell), Maximal strength (1-RM knee extension), Knee extension concentric power output (40-60-80% of 1RM, linear position transducer or encoder)
Shimizu et al. 2025 (38)	Squat	5 × 8	0.075	Pre-determined	2 sessions	2 minutes	Constant volume and intensity	8 weeks	100%	Countermovement jump height, squat jump height, drop jump height (20 cm and 40 cm) (jump mat); Maximum voluntary isometric knee extension strength (dynamometer); flywheel squat concentric and eccentric power (0.075 kg·m ² inertial disc, rotational sensor); Quadriceps heads' muscle thickness (ultrasound)
Spudić et al. 2024 (40)	Squat	3-5 × 10	0.025-0.25 0.175 (Author reported average via email)	Closest inertia to 70% of Pmax	2-3 sessions	5 minutes	Volume incremental and individualized intensity	8 weeks	100%	Countermovement jump height (force plates)
Xie et al. 2024 (70)	Squat	4 × 8	0.075	Pre-determined	2 sessions	3 minutes	Constant volume and constant intensity	6 weeks	90%	Maximal strength (1-RM back squat); Countermovement jump height and drop jump reactive strength index (jump mat)
Hu et al. 2024 (15)	Squat	4 × 9	0.05-0.112 0.081 (average)	Concentric velocity	2 sessions	3 minutes	Constant volume and individualized intensity	8 weeks	100%	Maximal strength (1-RM back squat); Countermovement and squat jump height (contact mat); 30 m sprint time (timing gates)
O'Brien et al. 2024 (39)	Flywheel Romanian deadlift	4 × 8	0.025-0.1 Set RPE method load allocation (8/10)	OPL based on the highest eccentric power	2 sessions	Unclear	Constant volume and individualized intensity	6 weeks	Unclear	Eccentric knee flexor strength (Hamstring kneeling/curling device); Maximal strength (3-RM barbell Romanian deadlift)

Asencio et al. 2024 (20)	Squat and horizontal lunge	2-4 × 6-11	Constant Inertia Flywheel Group: 0.08 Variable Inertia Flywheel Group: 0.12-0.08 0.10 (average)	Pre-determined	2 sessions	2 minutes	Volume incremental and constant intensity Volume incremental and intensity decremental	5 weeks	90%	Countermovement jump height (contact platform); Maximal strength (1-RM squat, linear encoder); 10 and 30 m sprint time (timing gates); 5-0-5 change of direction test time (timing gates); Flywheel squat concentric and eccentric power output (0.08 kg·m ² inertial disc, linear encoder)
Wang et al. 2024 (51)	Half-squat	3 × 10	0.18-0.31 0.245 (average)	Concentric velocity	3 sessions	1 minute	Constant volume and individualized intensity	8 weeks	100%	Quadriceps muscle thickness (B-mode ultrasonography); Maximal strength (1-RM back squat, optical encoder); Squat jump, countermovement jump height (contact platform)
Low et al. 2023 (71)	Half-squat and split squat	4 × 7	Ranges unclear RPE method load allocation (17/20)	Set RPE-based dynamic intensity adjustment	2 sessions	2 minutes	Constant volume and individualized intensity	4 weeks	Unclear	Countermovement jump height (force platform); Badcamp agility test time (switch trigger)
de Keijzer et al. 2023 (72)	Unilateral hip extension and leg curl	3-4 × 8	0.061-0.089 0.0725 (average)	Pre-determined	2 sessions	2 minutes	Volume undulated and intensity incremental	6 weeks	100%	Eccentric knee flexion strength at 60°/s ⁻¹ (isokinetic dynamometer); Flywheel unilateral knee flexion and hip extension concentric and eccentric power output (0.061 kg·m ² inertial disc, linear encoder)
Saez de Villareal et al. 2023 (73)	Lower Body: squat and leg curl	3 × 5-10	Unclear	Unclear	3 sessions	1.5 minutes	Constant volume and unclear intensity (lower body)	8 weeks	80%	Countermovement jump (electronic contact platform); 15 m sprint time (photoelectric cell)
Murton et al. 2023 (41)	Squat, Romanian Deadlift, and Bulgarian Split Squat	4-5 × 6-8	0.05	Pre-determined	2 sessions	Self-selected	Volume Incremental and constant intensity	4 weeks	100%	Countermovement jump height; squat jump height; drop jump reactive strength index (force platform)
Petre et al. 2023 (33)	Squat	4 × 8	0.025	Pre-determined	2 sessions	3 minutes	Constant volume and intensity	7 weeks	100%	Maximum voluntary isometric squat strength (axial load cell); Countermovement jump and squat jump height (optical measurement system)

Xu et al. 2023 (42)	Squat	4×7	Men: 0.11 Women: 0.075 0.0925 (average)	Pre-determined	2 sessions	2 minutes	Constant volume and intensity	4 weeks	100%	Concentric knee extension strength and power output at $240^\circ/\text{s}^{-1}$ (isokinetic dynamometer); Flywheel squat concentric power output ($0.05 \text{ kg}\cdot\text{m}^2$ inertial discs, rotational sensor); Countermovement jump height (jump mat)
Jarosz et al. 2023 (12)	Forward split-squat group: squat and forward split squat Lateral split squat group: squat and lateral split squat	Unclear	Squat: 0.1 Split squat variations: 0.05 0.075 (average)	Pre-determined	2 sessions	3 minutes (approx.)	Unclear volume and constant intensity	4 weeks	Unclear	5-0-5 change of direction test time (timing gate); Countermovement jump height (force plates); Maximal strength (1-RM back squat, linear transducer)
Puustinen et al. 2023 (74)	Bilateral squat, unilateral skating squat, leg curl, and leg press	$3\text{-}4 \times 6\text{-}7$	0.025-0.1 0.0625 (average)	OPL based on the highest average concentric power	1-2 sessions	3 minutes	Volume incremental and individualized intensity	8 weeks	Unclear	Countermovement jump height (infrared light mat); Flywheel squat concentric power output ($0.05 \text{ kg}\cdot\text{m}^2$ inertial disc, power encoder)
Filetti et al. 2022 (44)	Squat	5×6	0.061-0.081 0.071 (average)	Unclear	1-3 sessions	3 minutes	Constant volume and individualized intensity	6 weeks	Unclear	Rectus Femoris, vastus medialis, and vastus lateralis cross-sectional area (MRI)
Maroto-Izquierdo et al. 2022 (61)	Single-Leg Squat	4×7	0.05	Pre-determined	2 sessions	Unclear	Constant volume and intensity	6 weeks	100%	Thigh lean muscle mass (total body scanner); Maximal Strength (1-RM unilateral leg press); Leg press concentric power output (40-60-80% of 1RM, rotary encoder); Unilateral countermovement jump; squat jump; drop jump height (Contact platform)
Pecci et al. 2022 (21)	Squat	$3\text{-}4 \times 6\text{-}8$	0.025-0.05 0.0375 (average)	Pre-determined	2 sessions	2-3 minutes	Volume incremental and individualized intensity	6 weeks	75%	Concentric and eccentric extension strength at $60^\circ/\text{s}^{-1}$ (isokinetic dynamometer); Countermovement jump height (contact cells); 5+5 m shuttle run-sprint change of direction test time (single beam timing gate); 30 m sprint time (single beam timing gate)

Maroto-Izquierdo et al. 2022 (75)	Lateral raise, internal, and external rotation	4×7	0.001-0.113 Lateral raise: 0.025 Internal/External rotation: 0.074 0.057 (average)	OPL based on the highest concentric peak power	2 sessions	2 minutes	Constant volume and individualized intensity	6 weeks	100%	Concentric internal and external rotation strength and power output at 60° - 180° - $240^{\circ}/s^{-1}$ (isokinetic dynamometer); Shoulder muscle complex muscle thickness (B-mode ultrasonography)
Weng et al. 2022 (76)	Squat	4×7	0.06	Pre-determined	3 sessions	3 minutes	Constant volume and intensity	6 weeks	Unclear	Squat jump and countermovement jump height (force platform)
Morencos et al. 2022 (45)	Single-leg Romanian deadlift	$3-4 \times 10$	0.037	Pre-determined	2 sessions	2 minutes	Volume incremental and constant intensity	6 weeks	100%	Countermovement jump height (contact platform); 10 m sprint, 20 m sprint, and 30 m sprint test time (photoelectric cells); 5+5 m change of direction test time (photoelectric cells); Flywheel single-leg Romanian deadlift concentric and eccentric merged power output ($0.037 \text{ kg} \cdot \text{m}^2$ inertial disc, rotary encoder)
Asencio et al. 2022 (43)	Vertical flywheel group: squat and vertical split squat Horizontal flywheel group: lunge and horizontal split squat Mix flywheel group: squat, vertical split squat, lunge, and horizontal split squat	3×8	0.045-0.090 0.0675 (average)	Pre-determined	2 sessions	2 minutes	Constant volume and intensity decremental	4 weeks	Unclear	5-0-5 change of direction test time (contact platform); Countermovement jump height (contact platform); Maximal strength (1-RM back squat, linear position transducer)
Fousekis et al. 2021 (77)	Semi-squat	$3-4 \times 10$	Unclear	Unclear	2 sessions	2-3 minutes	Volume incremental and unclear intensity	6 weeks	Unclear	Concentric knee extension strength at 60 - 180 - $300^{\circ}/s^{-1}$ and eccentric knee extension strength at 60 - $150^{\circ}/s^{-1}$ (isokinetic dynamometer); Countermovement jump and squat jump height (jump mat);

										10m sprint, 30 m sprint, and Illinois agility test time (infrared photoelectric gates)
Martin et al. 2021 (46)	Upper Body: unilateral press, pull, push press, elbow flexion, and elbow extension Lower Body: squat and leg curl	3 × 10-15	0.06	Pre-determined	2 sessions	3 minutes	Volume incremental and constant intensity	18 weeks	100%	Countermovement jump height (infrared timing system); Fixed load muscle strength (50 kg bench press and full squat), (linear encoder); 10 m T swimming agility test time (photocell timing gate); 20 m sprint swim time (infrared timing systems)
Arsenis et al. 2021 (78)	Half-squat	3-6 × 5-6	0.0291-0.0978 0.0586 (average)	OPL based on the highest concentric mean power	2 sessions	3 minutes	Volume incremental and individualized intensity	8 weeks	Unclear	Concentric knee extension strength at 60-180-240°/s ⁻¹ (isokinetic dynamometer); Eccentric knee extension strength at 60-180°/s ⁻¹ (isokinetic dynamometer)
Timmins et al. 2021 (60)	Deadlift	Unclear	0.05-0.075 0.0625 (average)	Pre-determined	1-2 sessions	2 minutes	Volume and intensity incremental or decremental based on the cycle	39 weeks	73±10%	Muscle thickness, pennation angle, and fascicle length (B-mode ultrasound); Eccentric knee flexor strength (Nordic hamstring kneeling/curling device); Flywheel deadlift concentric and eccentric power output (flywheel ergometer); Maximum isometric strength (isometric mid-thigh pull, force plates); 40 m sprint time (timing gates)
Monajati et al. 2021 (56)	Double-leg squat, single-leg squat, straight-leg deadlift, leg curl, lunge, and hip extension	2 × 8	Squat device: 0.13 Conic pulley: 0.22 0.195 (average)	Pre-determined	2 sessions	2 minutes	Constant volume and intensity	6 weeks	100%	Concentric knee extension and flexion strength at 60°/s ⁻¹ (isokinetic dynamometer); Eccentric knee flexion strength at 60°/s ⁻¹ (isokinetic dynamometer)
Arsenis et al. 2020 (16)	Semi-squat	3-6 × 5-6	0.0291-0.0978 0.0586 (average)	OPL based on the highest concentric mean power	2 sessions	3 minutes	Volume incremental and individualized intensity	8 weeks	Unclear	Maximal strength (1-RM squat); Flywheel semi-squat concentric power output (0.0291-0.0978 kg*m-2 inertial discs, linear position transducer); Barbell semi-squat concentric power

										output (30-80% of 1-RM, linear position transducer); Squat jump, countermovement jump, and depth jump height (optical measurement system); 10 m and 30 m sprint time (light sensors)
O'Brien et al. 2020 (17)	Squat	4 × 10	0.075	Pre-determined	2 sessions	2 minutes	Constant volume and intensity	4 weeks	100%	Maximal strength (1-RM squat); Countermovement jump and squat jump height (contact platform); 10 m sprint time (timing gate); 5-0-5 change of direction test time (timing gate)
Presland et al. 2020 (79)	Unilateral leg curl	4-6 × 6-8	Conventional flywheel group: 0.050	Pre-determined	2 sessions	Unclear	Volume incremental and constant intensity	6 weeks	99.2%	Biceps femoris long head muscle thickness, pennation angle, and fascicle length (B-mode ultrasound); Maximum voluntary isometric knee flexion strength (load cell); Eccentric knee flexion strength (Nordic hamstring kneeling/curling device)
Cabanillas et al. 2020 (62)	Half-squat	4-6 × 10	0.0332	Pre-determined	1 session	3 minutes	Volume incremental and constant intensity	6 weeks	Unclear	Countermovement jump height (jump platform); 30 m sprint time (application)
Sagelv et al. 2020 (52)	Squat	3-4 × 7-9	0.025-0.1 0.0625 (average)	OPL based on dynamic concentric power /week, inertia increased if power produced >4 watts·kg ⁻¹ .	2 sessions	≥3 minutes	Volume undulating and individualized intensity	6 weeks	Unclear	10 m sprint time (single beam photocells); Countermovement jump height (portable force platform); Maximal strength (1-RM squat)
Coratella et al. 2019 (19)	Squat	4-6 × 8	0.11	Pre-determined	1 session	3 minutes	Volume incremental and constant intensity	8 weeks	Unclear	Countermovement jump and squat jump height (infrared device); T-test agility time (infrared device); 10 m and 30 m sprint time (infrared device); Concentric and eccentric knee extension at 60°/s ⁻¹ (isokinetic dynamometer); Eccentric knee flexion strength at 60°/s ⁻¹ (isokinetic dynamometer); Maximal strength (1-RM Squat);

										Lower-limb lean mass (total body scanner)
Sabido et al. 2019 (23)	Half squat	5 × 8	High inertia group: 0.075 Low inertia group: 0.025	Pre-determined	2 sessions	3 minutes	Constant volume and intensity	6 weeks	Unclear	Flywheel squat concentric and eccentric power output (0.025 and 0.075 kg·m ² inertial discs, power encoder); Maximal strength (1-RM Back Squat); Countermovement jump height (contact platform); 40 m sprint (photoelectric cells); T-test time (photoelectric cells)
Sanchez et al. 2019 (34)	Backward lunge, unilateral hamstring kicks, and half-squat	2-3 × 6	Conic pulley: 0.27 Squat device: 0.05 0.196 (average)	Pre-determined	2 sessions	1 minute	Volume incremental and constant intensity	5 weeks	Unclear	Countermovement jump height (electronic contact mat); Illinois agility test time (photoelectric cells)
Maroto-Izquierdo et al. 2019 (80)	Single-Leg Squat	4 × 7	0.05	Pre-determined	2 sessions	Unclear	Constant volume and intensity	6 weeks	100%	Thigh lean muscle mass (total body scanner); Maximal Strength (1-RM unilateral leg press); Leg press concentric power output (40-50-60-70-80-90% of 1RM, rotary encoder); Flywheel single-leg squat concentric and eccentric power output (0.05 kg·m ² inertia, rotary encoder); Bilateral and unilateral countermovement jump and squat jump height, (Contact platform)
Hernandez Davo et al. 2018 (81)	Bilateral flywheel group: bilateral half-squat Unilateral flywheel group: unilateral half-squat	4 × 8	0.025	Pre-determined	2 sessions	2 minutes	Constant volume and intensity	6 weeks	83.33%	Flywheel bilateral and unilateral squat concentric and eccentric power output (0.025 kg·m ² inertial disc, rotary encoder); Countermovement jump height (contact platform); T-test time (photoelectric cells)

Nunez et al. 2018 (14)	Lateral lunge flywheel group: lateral lunge Squat flywheel group: half-squat	4×7	Lateral lunge flywheel group: 0.05 Squat flywheel group: 0.1	Pre-determined	2 sessions	3 minutes	Constant volume and intensity	6 weeks	Unclear	Lower body muscles cross-sectional area (MRI); Countermovement jump height (infrared contact platform); 10-m sprint time (photoelectric cells); Change of direction test time (photoelectric cells); Flywheel squat and lateral lunge concentric and eccentric averaged power output (0.1 and 0.05 kg·m ² inertial discs, rotary encoder)
Illera-Domínguez et al. 2018 (13)	Bilateral half-squat	5×13	0.09	Pre-determined	2-3 sessions	3 minutes	Weekly volume undulating (training frequency) and constant intensity	4 weeks	Unclear	Flywheel squat concentric and eccentric force and power output (Chronoponic friction encoder); Maximum voluntary isometric knee flexion and extension strength (strain gauge); Lower body muscles volume (MRI); Quadriceps and hamstrings muscles cross-sectional area (MRI)
Skok et al. 2017 (82)	Squat flywheel group: bilateral squat Multidirectional flywheel group: unilateral flywheel exercises (backward lunges, defensive- like shuffling, side- step, crossover cutting, lateral crossover cutting, and lateral squat)	$6 \times 6-10$	0.27	Pre-determined	2 sessions	3 minutes	Volume incremental and constant intensity	8 weeks	85%	Change of direction tests time (photocells); 25-m sprint time (timing gates); Countermovement jump height (optical measurement system)
Sabido et al. 2017 (83)	Bilateral half-squat and lunge	$2-4 \times 8$	0.05	Pre-determined	1 session	2 minutes	Constant volume and intensity	7 weeks	90%	Flywheel squat and lunge concentric and eccentric power output (rotary encoder); Countermovement jump height (contact platform); 20 m sprint time (photoelectric cells);

										Maximal strength (1-RM back squat)
Maroto-Izquierdo et al. 2017 (22)	Leg Press	4×7	0.145	Pre-determined	2-3 sessions	3 minutes	Weekly volume undulating (training frequency) and constant intensity	6 weeks	100%	Maximal Strength (1-RM leg press); Leg press concentric power output (40-50-60-70-80-90% of 1-RM, rotary encoder); Vastus lateralis thickness (B-mode ultrasonography); Countermovement jump and squat jump height (contact platform); T-test time (photocell gates); 20 m sprint time (photocell gates)
Sanz-Lopez et al. 2017 (9)	Squat	4×7	0.07	Pre-determined	2 sessions	2 minutes	Constant volume and intensity	6 weeks	Unclear	Vastus lateralis pennation angle (lineal ultrasound transducer)
Gual et al. 2016 (47)	Squat	4×10	0.11	Pre-determined	1 session	2 minutes	Constant volume and intensity	24 weeks	Unclear	Countermovement jump height (contact mat); Flywheel squat concentric and eccentric power output ($0.11 \text{ kg} \cdot \text{m}^2$ inertial disc, rotary encoder)
Sanz-Lopez et al. 2016 (10)	Squat	4×7	0.07	Pre-determined	2 sessions	2 minutes	Constant volume and intensity	6 weeks	Unclear	Gastrocnemius medialis pennation angle (B-mode ultrasound)
De Hoyo et al. 2015 (84)	Horizontal first step acceleration	$5-7 \times 8$	0.1058	OPL based on maximum concentric power	3 sessions	3 minutes	Volume incremental and individualized intensity	6 weeks	Unclear	10-m and 20-m sprint time (electronic timing gates); Countermovement jump height (optical measurement system); 20-m zigzag change of direction test time (electronic timing gates); Maximum voluntary isometric knee extension strength (load cell)
Lundberg et al. 2014 (18)	Unilateral knee extension	4×7	0.11	Pre-determined	2-3 sessions	2 minutes	Weekly volume undulating (training frequency) and constant intensity	5 weeks	Unclear	Concentric knee extension strength at $30/60/120/180/210/240^\circ \cdot \text{s}^{-1}$ (isokinetic dynamometer); Maximum voluntary isometric knee extension strength (isokinetic dynamometer); Flywheel knee extension

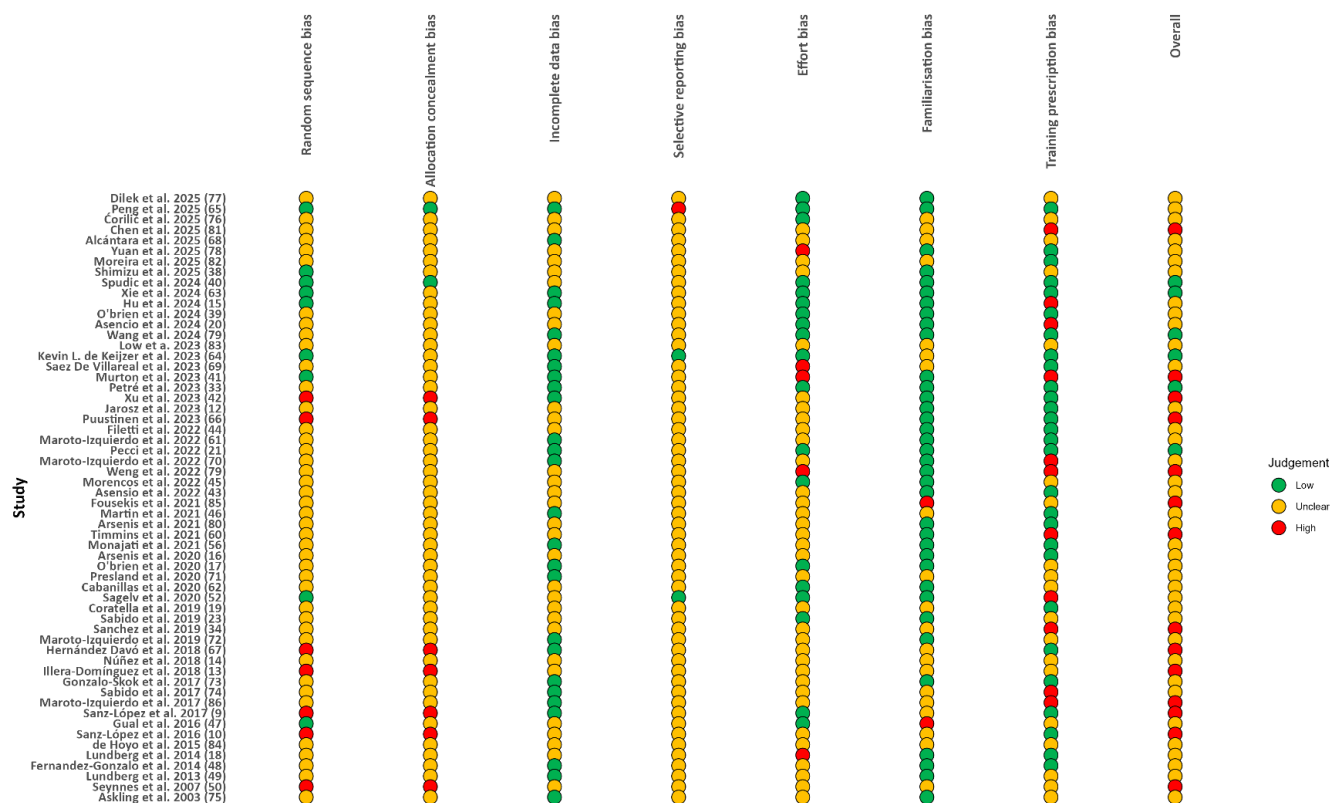
										concentric and eccentric strength and power output (flywheel ergometer); Quadriceps muscles volume and cross-sectional area (MRI)
Fernandez-Gonzalo et al. 2014 (48)	Supine squat	4×8	0.07	Pre-determined	2-3 sessions	Unclear	Weekly volume undulating (training frequency) and constant intensity	6 weeks	100%	Maximal strength (1-RM leg press); Squat jump, countermovement jump, and drop jump height (contact platform); Leg press concentric power output (40-50-60-70-80% of 1RM, rotary encoder); Thigh lean muscle mass (total body scanner)
Lundberg et al. 2013 (49)	Unilateral knee extension	4×7	0.11	Pre-determined	2-3 sessions	2 minutes	Weekly volume undulating (training frequency) and constant intensity	5 weeks	100%	Concentric knee extension strength at 30/60/120/180/210/240°.s ⁻¹ (isokinetic dynamometer); Maximum voluntary isometric knee extension strength (isokinetic dynamometer); Flywheel knee extension concentric strength and power output (flywheel ergometer); Quadriceps femoris muscle volume and cross-sectional area (MRI)
Seynnes et al. 2007 (50)	Bilateral knee extension	4×7	0.0758	Pre-determined	3 sessions	2 minutes	Constant volume and intensity	5 weeks	Unclear	Maximum voluntary isometric knee extension strength (isokinetic dynamometer); Vastus lateralis fascicle length and pennation angle (B-mode ultrasound); Quadriceps muscles cross-sectional area (MRI scanner)
C. Askling et al. 2003 (85)	Bilateral knee flexion	3×8	Unclear	Unclear	1-2 sessions	1 min (approx.)	Constant volume and unclear intensity	10 weeks	100%	Concentric and eccentric knee flexion strength at 60°.s ⁻¹ (kinetic communicator); 30 m sprint time (photocell gates)

RPE rated perceived exertion, 1-RM one-repetition maximum, Pmax maximum power (theoretical), OPL optimum power load(ing), 3-RM three-repetition maximum, MRI magnetic resonance imaging.

3.3. GRADE and Risk of Bias Assessment

The GRADE assessment indicated moderate-quality evidence for hypertrophy and eccentric power, low-quality evidence for overall strength, concentric power, overall jump, overall sprint, and COD, and very low-quality evidence for eccentric strength (Table 4).

Risk of bias judgments were based solely on information reported in the articles; no author contact was made for risk-of-bias-related clarification. Nine studies specified the randomization method used (15, 38, 40, 41, 47, 52, 64, 70, 72), while seven studies were rated with high risk because they either did not randomize group allocation or used single-group designs (9, 10, 13, 42, 50, 74, 81). The remaining studies did not report sufficient detail to allow a judgment on randomization. The seven studies that were identified as high risk in randomization were also rated high risk for allocation concealment bias, while only one study was rated low risk (40); the rest did not provide enough information to make a judgment. Twenty-six studies were judged low risk for incomplete data bias, as they reported drop-out reasons and adherence criteria (9, 15, 21, 22, 33, 41, 42, 46, 48, 49, 51, 52, 56, 61, 64, 67, 70, 72, 73, 75, 79, 80, 82, 83, 85). The rest either reported insufficient information or only partially addressed these criteria. Four studies reported preregistration; two provided accessible registrations that matched the published manuscripts (52, 72). Of the remaining two, one registration was inaccessible (47), and the other did not measure the prespecified outcomes and gave no rationale for this deviation; accordingly, it was judged at high risk of selective reporting bias (64). Nineteen studies were rated low risk for effort bias, as they explicitly stated that participants performed the concentric phase with maximal intent and were given at least two repetitions to build up momentum (9, 15, 17, 20, 23, 33, 39, 40, 45, 47, 51, 52, 62-65, 70, 72). Five studies were rated high risk because they provided no relevant information (18, 33, 41, 68, 76). The remaining studies reported only partial information. Thirty-five studies conducted ≥ 2 familiarization sessions (12, 15-18, 20, 21, 23, 33, 38-45, 48, 49, 51, 52, 56, 60-64, 68, 70, 74-76, 78, 80, 82, 85), 20 studies performed only one session (9, 10, 13, 14, 19, 22, 34, 46, 50, 65-67, 69, 71-73, 79, 81, 83, 84), and two studies did not report conducting any familiarization session (47, 77). The remaining studies did not report familiarization sessions. Eleven studies were identified as high risk for training prescription bias because they combined flywheel training with other resistance training exercises for the same muscle group or endurance training (15, 20, 34, 41, 52, 60, 66, 75, 76, 83, 86). Twenty-eight studies were rated low risk because they included only flywheel exercises in their training routines, either alongside sports training sessions or as a single exercise for the targeted muscle group (9, 10, 12, 16, 18, 19, 21, 33, 39, 40, 42-44, 46, 48, 51, 56, 61, 64, 68-70, 72-74, 78, 81, 82). The risk of bias assessment is illustrated in Figure 2.



3.4. Meta-Analytic Model Results

The results of the meta-analytic models, with and without moderators, are summarized in Tables 4 and 5. The impact of sensitivity analyses on the moderated models is described in detail below. Final model estimates are illustrated in Figures 3 and 4, which show the relationship between effect size and moderators. Forest plots for models without moderators, as well as initial model results with moderators, are available as supplementary material at osf.io/s3qgr/.

Table 4. Meta-Analytic Models' Results without Moderators

Outcome	<i>k</i>	<i>n</i>	SMC (95% CI)	Q	SE	t-value	Level 1 (<i>I</i> ² %)	Level 2 (<i>I</i> ² %)	Level 3 (<i>I</i> ² %)	Imprecision	Inconsistency	Risk of bias	Large magnitude of effect	Dose- response gradient	Quality of evidence
Overall Strength	69	433	0.648 (0.516- 0.781) ***	238.181	0.066	9.767	26.61	0	73.39	None	None	None	None	None	Low
Eccentric Strength	18	143	0.712 (0.465- 0.959) ***	57.743	0.117	6.081	23.65	13.09	63.26	None	-1	None	None	None	Very low
Concentric Power	73	263	0.943 (0.673- 1.212) ***	602.47	0.165	6.970	5.08	13.46	81.46	None	-1	None	+1	None	Low
Eccentric Power	31	191	1.058 (0.761- 1.354) ***	67.11	0.146	7.266	29.87	0	70.13	-1	None	None	+1	+1	Moderate
Hypertrophy	69	203	0.348 (0.258- 0.438) ***	142.147	0.045	7.707	28.37	2.76	68.87	None	None	None	None	+1	Moderate
Overall Jump	81	524	0.367 (0.278- 0.457) ***	203.492	0.045	8.165	31.64	0	68.36	None	None	None	None	None	Low

Overall Sprint	43	323	0.323 (0.170-0.477) ***	116.119	0.076	4.239	27.64	0	72.36	None	None	None	None	None	Low
Change of Direction	64	309	0.463 (0.244-0.681) ***	122.209	0.109	4.230	25.07	0	74.93	None	None	None	None	None	Low

*** $p < .001$, k total number of effect sizes, n number of analyzed participants, SMC standardized mean change, CI confidence intervals.

3.4.1. Overall Strength

Flywheel training moderately increased overall strength (Table 4). The initial meta-analytic model for overall strength identified weekly volume and bilateral limb involvement (compared to unilateral and mixed-limb protocols) as significant moderators ($\beta = 0.008$ [0.001, 0.016], $p = 0.031$, $\beta = -0.507$ [-0.882, -0.133], $p = 0.009$, and $\beta = -0.533$ [-0.955, -0.111], $p = 0.014$ respectively). Sensitivity analyses revealed seven influential studies, and their removal changed the model interpretation: weekly volume was no longer statistically significant ($\beta = 0.005$ [-0.002, 0.012], $p = 0.164$), while the effect favoring bilateral exercises became more pronounced compared to unilateral ($\beta = -0.674$ [-0.949, -0.400], $p < .001$) and mixed-limb protocols ($\beta = -0.571$ [-0.945, -0.198], $p = 0.004$) (Table 5.). The removal of influential studies explained 64.96% of within-study heterogeneity (τ^2) and 78.70% of between-study heterogeneity (τ^2).

A secondary analysis showed no significant effects: lower vs. higher volume ($\beta = -0.167$, 95% CI [-0.479, 0.145], $p = 0.288$) and lower vs. higher inertia ($\beta = 0.145$, 95% CI [-0.155, 0.439], $p = 0.344$).

3.4.2. Eccentric Strength

Flywheel training moderately increased eccentric strength (Table 4). Due to limited data, weekly volume and inertia were not analysed within the same model, and neither was found to be statistically significant (Table 5). Sensitivity analyses identified two influential studies; however, their removal did not alter the non-significant findings for weekly volume ($\beta = -0.010$ [-0.021, 0.001], $p = 0.067$). The weekly volume model excluding influential studies explained 51.29% of within-study τ^2 and 25.17% of between-study τ^2 .

A secondary analysis found no significant effects: lower vs. higher volume ($\beta = 0.015$, 95% CI [-0.514, 0.545], $p = 0.952$).

3.4.3. Concentric Power

Flywheel training revealed a large increase in concentric power (Table 4). The initial meta-analytic model showed that periodization and the power assessment method were significant moderators ($\beta = -1.140$ [-2.263, -0.016], $p = 0.047$, and $\beta = -0.922$ [-1.190, -0.653], $p < 0.001$, respectively). The sensitivity analyses revealed six influential studies. After their removal, the model's interpretation changed: a statistically significant difference was no longer present in periodization ($\beta = -0.865$ [-1.952, 0.223], $p = 0.117$). However, the power assessment method remained a significant moderator, and participant training level also emerged as significant, indicating that flywheel training has a smaller effect in trained athletes compared to the highly trained population (Table 5). The model excluding influential studies explained 15.61% of within-study τ^2 and 84.18% of between-study τ^2 .

A secondary analysis found no significant differences: lower vs. higher volume ($\beta = 0.255$, 95% CI [-0.342, 0.852], $p = 0.397$) and lower vs. higher inertia ($\beta = 0.226$, 95% CI [-0.367, 0.819], $p = 0.450$).

3.4.4. Eccentric Power

Flywheel training revealed a large increase in eccentric power (Table 4). The initial meta-analysis model showed that no model moderators were significant (Table 5). The sensitivity analyses identified three influential studies.

After their removal, the model's interpretation changed; the volume-inertia interaction and inertia moderators were found to be significant ($\beta = -0.727$ [-1.437, -0.017], $p = 0.045$ and $\beta = 50.926$ [1.467, 100.386], $p = 0.044$, respectively). The model excluding influential studies explained 86.23% of within-study τ^2 and 75.40% of between-study τ^2 .

A secondary analysis revealed no significant differences: lower vs. higher volume ($\beta = 0.247$, 95% CI [-0.506, 1.000], $p = 0.508$) and lower vs. higher inertia ($\beta = -0.011$, 95% CI [-0.790, 0.769], $p = 0.978$).

3.4.5. Hypertrophy

Flywheel training showed a small increase in hypertrophy (Table 4). The initial meta-analytic model revealed that weekly volume was a significant moderator ($\beta = 0.012$ [0.001, 0.023], $p = 0.035$) (Table 5). The sensitivity analyses revealed eight influential studies. After their removal, weekly volume remained a significant moderator ($\beta = 0.016$ [0.003, 0.029], $p = 0.018$). The model excluding influential studies explained 15.54% of within-study τ^2 and 8.55% of between-study τ^2 .

A secondary analysis found no significant differences: lower vs. higher volume ($\beta = -0.154$, 95% CI [-0.385, 0.078], $p = 0.190$) and lower vs. higher inertia ($\beta = 0.071$, 95% CI [-0.122, 0.265], $p = 0.463$).

3.4.6. Overall Jump

Flywheel training showed a small increase in overall jump performance (Table 4). The initial meta-analytic model for overall jump performance showed no statistically significant effects for any of the included moderators (Table 5). However, the sensitivity analyses revealed that eight studies were influential. After their removal, the results remained unchanged. The model excluding influential studies explained 43.99% of within-study τ^2 and 86.33% of between-study τ^2 .

A secondary analysis found no significant differences: lower vs. higher volume ($\beta = 0.029$ [-0.150, 0.208], $p = 0.747$) and lower vs. higher inertia ($\beta = 0.018$ [-0.157, 0.193], $p = 0.839$).

3.4.7. Overall Sprint

Flywheel training showed a small increase in overall sprint performance (Table 4). The initial meta-analytic model showed that no statistically significant differences were observed for model moderators (Table 5). Sensitivity analyses identified one influential study; however, its removal did not change the interpretation of the results. The model excluding the influential study showed no change in explained within-study τ^2 and between-study τ^2 .

A secondary analysis found no significant differences: lower vs. higher volume ($\beta = 0.239$ [-0.129, 0.607], $p = 0.196$) and lower vs. higher inertia ($\beta = -0.354$ [-0.723, 0.015], $p = 0.060$).

3.4.8. Change of Direction

Flywheel training revealed a small increase in change of direction performance (Table 4). The initial meta-analytic model did not show a significant difference in any of the model moderators (Table 5). Sensitivity analyses

identified four influential studies. After their removal, the results remained unchanged. The model without influential studies explained 31.02% of between-study τ^2 , while the within-study τ^2 remained unchanged.

A secondary analysis found no significant differences: lower vs. higher volume ($\beta = 0.252$ [-0.322, 0.826], $p = 0.382$) and lower vs. higher inertia ($\beta = 0.082$ [-0.509, 0.674], $p = 0.781$).

Table 5. Final Meta-analytic Models' Results with Moderators

Outcome	Moderator	Studies	Effect Sizes	β (95% CI)	SE	t-value	p-value	Overall moderator effect (Omnibus test)	R^2 (level 2)	R^2 (level 3)
Overall Strength	Intercept ^a	31	56	0.544 (-0.112, 1.200)	0.326	1.669	0.102	$F(8, 47) = 3.917, p < 0.001$	64.96	78.70
	Volume/week	31	56	0.005 (-0.002, 0.012)	0.003	1.414	0.164			
	Inertia	31	56	-2.244 (-10.229, 5.741)	3.969	-0.565	0.575			
	Recreationally active	31	56	0.199 (-0.130, 0.527)	0.163	1.216	0.230			
	Trained	31	56	-0.041 (-0.344, 0.262)	0.151	-0.275	0.785			
	Bilateral and Unilateral	31	56	-0.571 (-0.945, -0.198)	0.186	-3.075	0.004**			
	Unilateral	31	56	-0.674 (-0.949, -0.400)	0.136	-4.951	< .001***			
	Periodization, yes	31	56	0.072 (-0.279, 0.423)	0.174	0.414	0.681			
	Volume/week: inertia	31	56	0.003 (-0.073, 0.080)	0.038	0.087	0.931			
Eccentric Strength	Intercept	10	16	1.280 (0.575, 1.985)	0.329	3.892	0.002	$F(1, 14) = 3.928, p = 0.067$	51.29	25.17
	Volume/week	10	16	-0.010 (-0.021, 0.001)	0.005	-1.982	0.067			
Concentric Power	Intercept ^b	19	67	2.601 (-2.162, 7.364)	2.379	1.094	0.279	$F(9, 57) = 1.905, p = 0.070$	15.61	84.18
	Volume/week	19	67	0.004 (-0.046, 0.054)	0.025	0.162	0.872			
	Inertia	19	67	-0.994 (-62.399, 60.411)	30.665	-0.032	0.974			

	Recreationally active	19	67	-0.629 (-1.717, 0.458)	0.543	-1.159	0.251			
	Trained	19	67	-1.858 (-3.567, -0.148)	0.854	-2.176	0.034*			
	Bilateral and Unilateral	19	67	-1.201 (-2.894, 0.491)	0.845	-1.421	0.161			
	Unilateral	19	67	-0.050 (-0.770, 0.670)	0.360	-0.138	0.890			
	Periodization, yes	19	67	-0.865 (-1.952, 0.223)	0.543	-1.593	0.117			
	Power assessment method, traditional	19	67	-0.918 (-1.435, -0.401)	0.258	-3.554	< .001***			
	Volume/week: inertia	19	67	-0.041 (-0.859, 0.777)	0.409	-0.099	0.921			
Eccentric Power	Intercept ^c	13	30	-2.400 (-6.177, 1.376)	1.830	-1.312	0.202	$F(5, 24) = 1.709, p = 0.171$	86.23	75.40
	Volume/week	13	30	0.050 (-0.005, 0.104)	0.026	1.889	0.071			
	Inertia	13	30	50.926 (1.467, 100.386)	23.964	2.125	0.044*			
	Bilateral and Unilateral	13	30	0.236 (-0.410, 0.883)	0.313	0.755	0.457			
	Unilateral	13	30	-0.254 (-0.653, 0.144)	0.193	-1.316	0.201			
	Volume/week: inertia	13	30	-0.727 (-1.437, -0.017)	0.344	-2.114	0.045*			
Hypertrophy	Intercept ^d	14	60	-0.790 (-1.823, 0.243)	0.515	-1.533	0.131	$F(5, 54) = 1.887, p = 0.112$	15.54	8.55
	Volume/week	14	60	0.016 (0.003, 0.029)	0.006	2.446	0.018*			
	Inertia	14	60	12.368 (-0.918, 25.653)	6.627	1.866	0.067			
	Unilateral	14	60	-0.028 (-0.232, 0.176)	0.102	-0.273	0.786			

	Periodization, yes	14	60	-0.174 (-0.419, 0.070)	0.122	-1.428	0.159			
	Volume/week: inertia	14	60	-0.159 (-0.322, 0.004)	0.081	-1.953	0.056			
Overall Jump	Intercept ^e	38	71	-0.125 (-0.861, 0.612)	0.367	-0.339	0.736	$F(10, 54) = 0.755, p = 0.670$ 43.99 86.33		
	Volume/week	38	71	0.005 (-0.003, 0.013)	0.004	1.347	0.184			
	Inertia	38	71	8.331 (-1.446, 18.108)	4.877	1.708	0.093			
	Recreationally active	38	71	-0.154 (-0.452, 0.145)	0.149	-1.033	0.306			
	Trained	38	71	0.008 (-0.240, 0.256)	0.124	0.067	0.947			
	Bilateral and Unilateral	38	71	-0.051 (-0.291, 0.392)	0.170	0.298	0.767			
	Unilateral	38	71	0.139 (-0.136, 0.414)	0.137	1.015	0.314			
	Periodization, yes	38	71	0.097 (-0.144, 0.338)	0.120	0.807	0.423			
	Physiological function, only concentric (squat jump)	38	71	0.014 (-0.257, 0.285)	0.135	0.105	0.917			
	Physiological function, only concentric (countermovement jump)	38	71	-0.023 (-0.278, 0.233)	0.127	-0.179	0.859			
	Volume/week: inertia	38	71	-0.089 (-0.196, 0.018)	0.053	-1.672	0.100			
Overall Sprint	Intercept ^f	23	37	0.026 (-1.438, 1.489)	0.718	0.036	0.972	$F(5, 31) = 0.655, p = 0.660$ 0 0		
	Volume/week	23	37	-0.000 (-0.019, 0.018)	0.009	-0.028	0.978			

	Inertia	23	37	5.770 (-13.255, 24.794)	9.328	0.619	0.541			
	Periodization, yes	23	37	0.131 (-0.309, 0.570)	0.215	0.608	0.548			
	Sprint distance, shorter than 20 m	23	37	-0.006 (-0.231, 0.219)	0.110	-0.051	0.960			
	Volume/week: inertia	23	37	-0.041 (-0.251, 0.170)	0.103	-0.392	0.697			
Change of Direction	Intercept ^g	19	49	0.856 (-3.081, 4.793)	1.951	0.438	0.663	$F(6, 42) = 0.840, p = 0.546$	0	31.02
	Volume/week	19	49	-0.007 (-0.054, 0.040)	0.023	-0.299	0.766			
	Inertia	19	49	12.061 (-31.119, 55.242)	21.397	0.563	0.576			
	Bilateral and Unilateral	19	49	-0.427 (-1.678, 0.824)	0.620	-0.689	0.494			
	Unilateral	19	49	-0.312 (-1.249, 0.625)	0.464	-0.671	0.505			
	Periodization, yes	19	49	0.479 (-0.617, 1.576)	0.543	0.882	0.382			
	Volume/week: inertia	19	49	-0.133 (-0.591, 0.325)	0.227	-0.585	0.561			

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ^ahighly trained, bilateral, and periodization (no) parameters were added to the intercept; ^bhighly trained, bilateral, periodization (no), and power assessment method (flywheel) parameters were added to the intercept; ^cthe bilateral parameter was added to the intercept; ^dbilateral and periodization parameters were added to the intercept, ^ehighly trained, bilateral, periodization (yes), fast stretch-shortening cycle (SSC) parameters were added to the intercept; ^fperiodization (no) and sprint distance (longer than 20m) were added to the intercept; ^gbilateral and periodization (no) parameters were added to the intercept.

4. Discussion

This systematic review with meta-analysis investigated the longitudinal effects of flywheel training and examined the dose-response relationship on the magnitude of adaptation in muscle strength, power, hypertrophy, and the performance of athletic tasks. The findings can be summarized as follows: 1) Flywheel training produced large improvements in power (concentric and eccentric), moderate improvements in strength (overall and eccentric), and small improvements in hypertrophy and performance of athletic tasks: jump, sprint, and COD; 2) A dose-response relationship was observed only between weekly training volume and hypertrophy; 3) Bilateral exercises produced greater overall strength gains than unilateral or mixed exercise limb involvement; 4) Concentric power adaptations were more pronounced when measured with flywheel-specific assessments compared to traditional power assessment methods; 5) Flywheel training was more effective for highly trained athletes than for trained athletes in improving concentric power; 6) Higher inertial loads led to greater improvements in eccentric power; 7) A negative volume-inertia interaction effect was observed for eccentric power, indicating that high training volumes should not be combined with high inertial loads; and 8) No statistically significant dose-response relationship was found between weekly training volume or inertia and overall strength, eccentric strength, concentric power, jump, sprint, or COD. Taken together, these findings imply that while flywheel training is a broadly effective modality for enhancing neuromuscular performance, the magnitude of adaptation is highly context-dependent, relying heavily on exercise specificity, athlete training status, and—specifically for hypertrophy and eccentric power—precise manipulation of the training dose.

4. 1. The Effects of the Flywheel Training on Strength, Power, and Hypertrophy

Strength

Flywheel training resulted in a moderate increase in both overall and eccentric strength. This contrasts with Petré et al. (24) who reported a large increase ($SMD = 1.33$) in maximal strength. The smaller effects observed may reflect differences in effect size calculation methods and the studies included in each analysis. Given the inclusion of more recent studies in the current review, it appears that the magnitude of flywheel training effectiveness for strength development may be smaller than previously thought.

Additionally, when comparing flywheel training to traditional resistance training, the literature is mixed. One meta-analysis found a moderate increase favoring flywheel training ($SMD = 0.66$, $p < 0.001$) (86), whereas another study reported no significant difference between the two training methods ($SMD = 0.24$, $p = 0.35$) (87). Strength improvements from flywheel training likely stem from morphological and increased neural demands (88), mechanisms that are similar to those of other resistance training modalities (89). Moreover, a plausible explanation for these improvements is the eccentric overload provided by flywheel training, although a previous meta-analysis indicated that flywheel-related force adaptations occurred both with and without eccentric overload (90). Consequently, the specific contexts and mechanisms by which flywheel training may or may not be superior to traditional methods require further investigation.

Our results revealed that weekly training volume was not a significant moderator, meaning higher weekly repetitions did not lead to greater improvements in overall strength, primarily reflecting concentric adaptations. This finding is supported by current resistance training literature. For example, Jukic et al. (31) showed that higher

training volumes (higher velocity loss thresholds) did not stimulate greater maximal strength adaptations. Similarly, Robinson et al. (91) found that higher training volumes (tasks to failure) were not necessary for strength adaptation. Furthermore, Clemente et al. (7) reported no difference between lower and higher volumes in improving strength ($g = 0.078$, $p = 0.689$). In contrast, Ralston et al. (40) found that moderate (5-9 sets) and high (≥ 10 sets) weekly volumes were more effective than lower volume training ($p = 0.003$ and $p = 0.04$, respectively), though the effect sizes were trivial (MD: 0.18 and 0.15). These findings suggest concentric strength adaptations may not be highly sensitive to volume ranges. The lack of a volume effect may reflect the necessary neural and morphological changes (89) that occur even with lower training doses. The inherent high-intensity stimulus of flywheel training, particularly during the eccentric phase (88), may provide sufficient adaptation, making additional volume unnecessary. Further research is warranted to clarify the specific physiological drivers of this low-volume efficiency.

Regarding exercise allocation, our review suggests bilateral flywheel exercises are more effective for improving overall strength than unilateral or mixed-limb approaches. This may partly reflect testing specificity, as most flywheel studies assess bilateral strength. Another possible reason might be that bilateral resistance exercises induce greater agonist neuromuscular activation (92, 93), which could contribute to superior strength adaptations. In support of this, Liao et al. (92) showed that bilateral limb allocation statistically improved bilateral strength (SMD = 0.43, $p = 0.004$), whereas unilateral limb allocation did not significantly improve unilateral strength (SMD = 0.26, $p = 0.07$). Furthermore, unilateral exercises may impose greater stability demands that limit force production, thereby limiting maximal agonist muscle adaptation (92, 93). Therefore, unilateral exercises may be beneficial for improving stability-related strength qualities but not maximal strength. Unfortunately, the available flywheel studies that directly compared the effectiveness of bilateral and unilateral exercises did not include any strength testing in both bilateral and unilateral movement patterns (14, 81, 82). Therefore, future studies should investigate whether there are differences in adaptational responses to strength between bilateral, unilateral, and mixed exercise allocations.

Finally, no dose-response relationship was observed for eccentric strength. Eccentric strength may be less sensitive to volume changes, possibly because the eccentric overload characteristics of flywheel training provide a sufficient stimulus regardless of volume. Although both the presence and absence of eccentric overload improved force (90). However, adaptation was not specified for strength (90). Similar trends are observed in other eccentric resistance training modalities: low and high doses of Nordic hamstring curls and stiff-leg deadlifts (10 vs. 40 repetitions/week) produced similar eccentric strength gains (94). Likewise, a meta-analysis reported similar effects for low and high training volumes ($g = 2.12$ and $g = 2.28$, respectively) (95). These collective findings reinforce the notion that for both concentric and eccentric strength, intensity and mechanical stimulus appear to be more critical drivers of adaptation than accumulated training volume.

Power

The power findings of the current review are consistent with Petré et al. (24), who reported a large increase from flywheel training in concentric power development (SMD = 1.19). These results indicate that flywheel training is highly effective for enhancing concentric power. Comparisons to traditional resistance training, however, are

mixed. For example, a previous study found only a small increase in favor of flywheel training (SMD = 0.47) (87). This smaller effect may reflect the use of less flywheel-specific assessment techniques, such as countermovement jump performance, which may not capture flywheel-specific power adaptation. This highlights the task-specific nature of power adaptations and raises questions about the validity of using countermovement jumps for assessing power changes. At the very least, it is a limitation when assessing true maximal power output (96). Previous research suggests that normalizing power to body mass measures the same construct as jump height (97, 98). Our findings also show the importance of the specificity of the assigned power test. In our study, flywheel-based power test adaptation was superior compared to traditional methods. Therefore, specific flywheel power tests, rather than including weight stacks and jump-derived power data, should be prioritised. Additionally, the superiority of flywheel training for power development compared to traditional resistance training remains unclear.

Our model results for concentric power indicated that flywheel training was more effective in highly trained athletes compared to trained individuals ($p = 0.034$), whereas a similar trend was not observed in recreationally active populations ($p = 0.251$). Since concentric effort depends on the user's intention and capacity (24), highly trained athletes may generate greater power and thus benefit more from flywheel training. However, this assumption is not fully supported by the literature. Ascencio et al. (99) found no difference between stronger and weaker athletes in concentric or eccentric power across various inertias, though stronger athletes could achieve eccentric overload across all inertias, while weaker athletes did so only at higher inertias ($0.075\text{--}0.1\text{ kg}\cdot\text{m}^2$) (99). To further investigate, we included the training level moderator in our eccentric power trial models, but did not observe the same trend. This suggests that competition level (trained vs. highly trained) may not be directly related to strength level in the included flywheel studies. However, the limited number of effect sizes for eccentric power prevented us from assessing the effects of moderators within the same model. Therefore, further research is needed to clarify the relationship between athlete training level (both strength and competition) and concentric and eccentric power generation.

For eccentric power, both inertia and volume-inertia interaction were significant moderators. Our inertial range ($0.025\text{--}0.11\text{ kg}\cdot\text{m}^2$) aligned with Ascencio's findings (99), showing that participants could produce eccentric overload at higher inertias. However, high training volumes combined with high inertias may attenuate adaptations, highlighting the importance of carefully managing both variables. In traditional resistance training, Naclerio et al. (100) reported that lower training volumes were more effective for improving squat power, whereas bench press improvements favored moderate and high volumes (100). Natera et al. (4) suggested that 3–5 sets of 10–20 repetitions at 30–40% 1-RM enhance repeated power, but high volumes can induce fatigue—a mechanism that may also apply to flywheel training. Thus, training volume should be carefully prescribed to optimize eccentric power adaptations.

Interestingly, no relationship was observed between inertia and concentric power development in our study within the inertial ranges of $0.025\text{--}0.15\text{ kg}\cdot\text{m}^2$. Similarly, McErlain-Naylor et al. (101) found a non-linear relationship between inertia and power within an identical range ($0.029\text{--}0.121\text{ kg}\cdot\text{m}^2$). This suggests that the concept of optimal inertia for power output in flywheel training is highly individualized, and optimal power inertia can be

achieved at different inertias. Therefore, narrow inertial ranges may not capture an individual's optimal power inertia, and this limitation at the study level could obscure the relationship between inertia and concentric power. Moreover, Asencio et al. (99) reported that peak concentric and eccentric power generation depend on individual strength levels and body mass, with weaker athletes showing visually varied power patterns, while both stronger and weaker groups achieved peak power at lower inertias. On the contrary, Spudis et al. individualized inertia based on 70% of participants' Pmax values and found that these values were higher than those used in previous studies (with an average group inertia of 0.175 kg·m²) (40). This indicates that optimal inertia for power varies across studies and individuals. Again, these results suggest that power generation is highly individual, and optimal power inertia prescription should be performed using power profiling or velocity-threshold methods across a broad inertial spectrum. Such an approach may improve understanding of the relationship between inertia and concentric power adaptations.

Hypertrophy

In contrast to our finding of a small overall effect of flywheel training, previous reviews reported a moderate increase in muscle cross-sectional area and muscle volume/mass (SMD = 0.59) in pre-post comparisons (24) and compared to traditional resistance training (SMD = 0.59) (86). Again, these discrepancies may be due to differences in the studies included and the incorporation of more recent research in the present review. Also, these findings indicate that flywheel training may not be as effective for improving muscle hypertrophy as previously supposed, despite its eccentric overload characteristic (22, 86, 102). One study even showed that muscle volume improvements with flywheel resistance training can occur independently of eccentric overload (90). Future studies should further investigate the role of eccentric overload and directly compare the effects of flywheel and traditional resistance training on muscle hypertrophy.

Our analysis revealed a significant relationship between weekly training volume and the magnitude of adaptation. These findings are consistent with results from various resistance training studies (3, 6, 31, 91, 103). Physiologically, the recruitment of both type I and type II muscle fibers through motor unit activation leads to mechanical tension production (104). The mechanical tension produced stimulates anabolic pathways and muscle growth (104), particularly in type II fibers (91). Notably, training with loads of 40–80% of 1-RM and performing repetitions close to failure is known to maximize mechanical tension and motor unit recruitment (91, 105). Given these physiological mechanisms, a key limitation of flywheel training is that repetitions are often not performed near failure, which may partly explain the small pre-post hypertrophy effects observed in this review. Moreover, although repetitions are not performed close to failure with flywheel training, increasing flywheel training volume (sets and repetitions)—bringing participants closer to failure—may enhance hypertrophic responses. Furthermore, increased volume may induce greater metabolic stress, potentially leading to enhanced muscle growth. For example, studies have found that flywheel protocols produce higher acute lactate levels compared to traditional resistance training (106, 107). Accumulated lactate levels, as a by-product of high-intensity exercise, may stimulate hypertrophy signaling pathways (104). Additionally, it was shown that muscle oxygenation saturation decreased to a greater degree after flywheel squats compared to traditional squats (107). Given the metabolic demands of flywheel training (11), it is plausible that these metabolic stress-related factors contribute to hypertrophy adaptations in flywheel training. Thus, performing higher intra-set repetitions and increasing the

number of sets in flywheel training can enable users to achieve higher total repetition training volumes and bring them closer to muscle failure. This approach will also promote greater mechanical fatigue (108) and potentially generate greater metabolic stress. However, these hypotheses require further investigation.

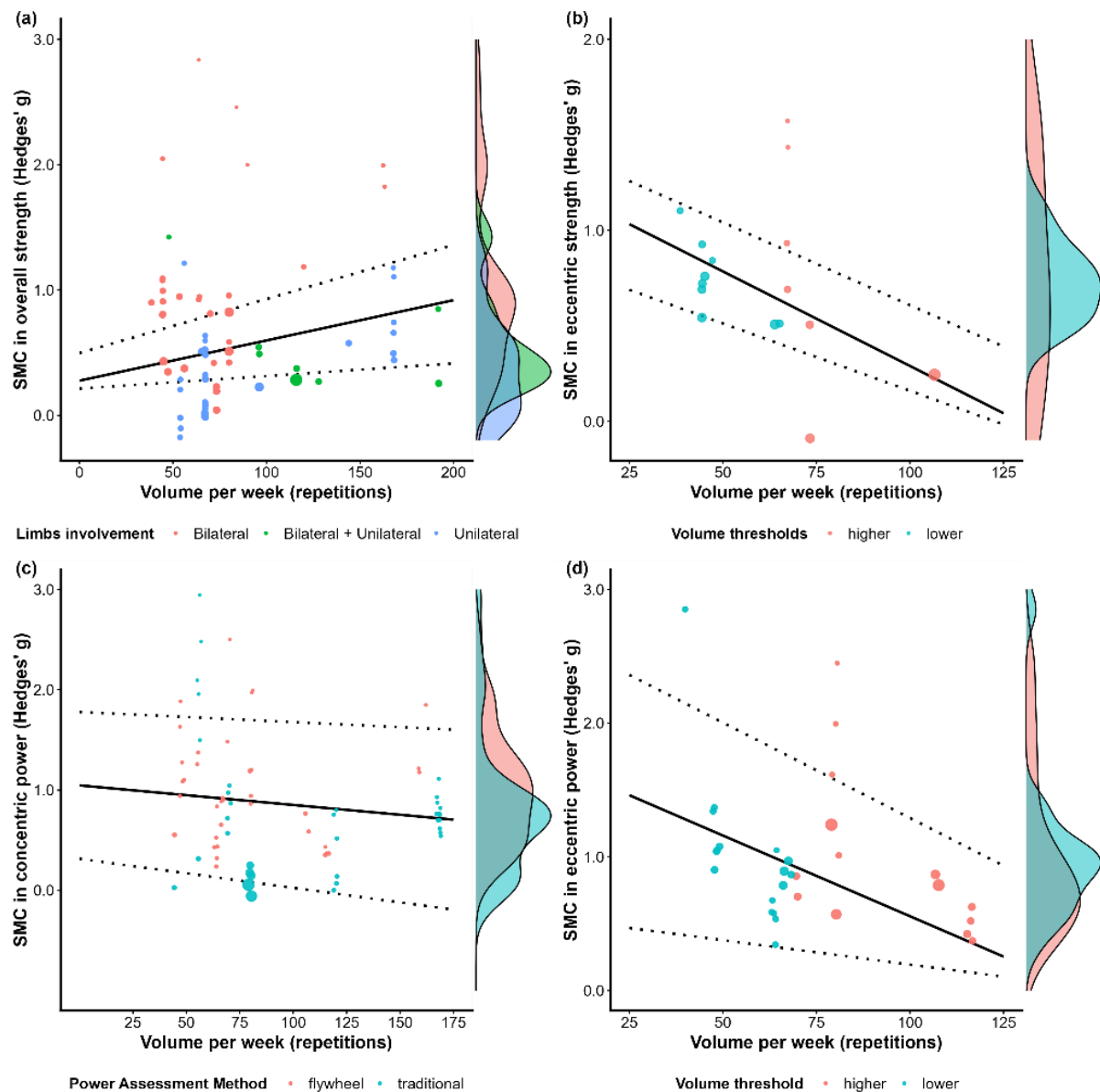


Figure 3. The moderating effects of (a) weekly training volume on overall strength, (b) eccentric strength, (c) concentric power, and (d) eccentric power, visualized from the final multilevel meta-regression models. The plotted relationships are adjusted for all moderators in the model (a) (inertia, training level, limb involvement, periodization, and volume-inertia interaction), (b) (only weekly training volume), (c) (inertia, training level, limb involvement, periodization, power assessment method, and volume-inertia interaction), and (d) (inertia, limb involvement, and volume-inertia interaction). Solid lines represent the estimated effect, and dotted lines show the 95% confidence interval. Points are weighed by the inverse of their variance.

4.2. The Effects of the Flywheel Training on the Performance of Athletic Tasks

Our analyses indicate that flywheel training elicits a small increase in athletic task performance (jump, sprint, and COD). In contrast, Petré et al. (24) reported a large pre-post increase from flywheel training on vertical jump (SMD = 0.85), while Maroto-Izquierdo et al. (86) observed a small increase favoring flywheel training compared to traditional resistance training (SMD = 0.46). Additionally, Shimizu et al. (109) found that flywheel training produced a large increase versus a control group (SMD = 0.81), but only a small increase compared to traditional resistance training (SMD = 0.10). For sprint performance, Petré et al. found a large pre-post increase from flywheel training (ES = 1.00). Another review (86) reported a small increase favoring flywheel training over traditional resistance training (ES = 0.41). These discrepancies likely stem from the limited number of studies included in previous reviews (24, 109), which reduced the precision of estimates. In contrast, the current review incorporates a substantially larger pool of effect sizes, resulting in more precise findings. For COD performance, our findings showed a small increase in flywheel training (ES = 0.47), whereas a prior review (25) reported a large increase (ES = 1.63). The latter review relied on substantial data from paediatric populations (6/10 studies, average age = < 18 years old), which may partially explain the large increase observed (110). Again, the small number of available studies may have contributed to variation across reviews. The consistently small pre-post improvements in the current review may reflect the limited transfer of flywheel resistance training to the rapid force production demands of athletic tasks. Similar limitations in transferability between resistance training and fast, dynamic multi-joint movements were highlighted previously (111). Therefore, to maximize performance outcomes, integrating flywheel training with movement-pattern-specific exercises may enhance the translation to athletic tasks.

In our study, we did not observe a dose-response relationship for jump, sprint, or COD. This is not entirely unexpected given the lack of consensus in the broader literature regarding how resistance training volume influences the performance of athletic task adaptation. For example, similar to our findings, Clemente et al. (7), also reported no clear dose-response effect when comparing lower and higher training volumes. They found that neither volume condition were more effective for improving vertical jump ($g = 0.12$, $p = 0.451$), vertical jump after a drop jump ($g = -0.20$, $p = 0.468$), 10-m sprint ($g = -0.20$, $p = 0.224$), 20–40-m sprint ($g = 0.025$, $p = 0.903$), or COD ($g = -0.036$, $p = 0.845$). The authors noted heterogeneity in training volumes and differences between lower and higher training volumes of the included studies, making it difficult to generalize these findings. The study also suggested that lower volumes were as effective as higher volumes in providing the required stimulus for athletic task adaptation.

Conversely, Jukic et al. (31) found that lower volumes (lower velocity loss thresholds) were more effective for improving countermovement jump and sprint performance. They argued that higher velocity loss thresholds (higher volumes) may induce excessive fatigue and hinder performance in high-velocity athletic tasks. Particularly, given the eccentric biased nature of flywheel training, excessive eccentric fatigue can reduce the effectiveness of SSC (112), which is essential for jump, sprint, and COD (113-115). An acute study supports this idea, showing a larger decrease in countermovement jump height at higher-velocity loss thresholds (10% and 15%) compared with 5% (116). However, such acute responses may not reflect longitudinal adaptations. If higher volumes impaired performance in our review, greater decrements would be expected, yet this was not observed.

Similar to Clemente et al. (7), it appears that lower volumes may provide sufficient stimulus for athletic performance adaptations in flywheel training, potentially by helping maintain higher intra-set velocities relevant to jump, sprint, and COD tasks.

Surprisingly, we did not observe any relationship between inertia and the performance of athletic tasks. For jump performance, a dose-response favoring lower inertias was expected, given that our dataset largely included countermovement and drop jumps (slow- and fast-SSC actions). One study showed that low loads allowed performing a better SSC function by reducing the transition from eccentric to concentric phase (117). A previous flywheel study reported that lower inertias ($0.025\text{--}0.0375\text{ kg}\cdot\text{m}^2$) produced higher stretch-shortening utilization ratios than higher inertias, although with substantial inter-individual variability (118). Similar findings in traditional resistance training indicate that lighter loads ($\sim 30\%$ 1RM) better improve jump performance by enabling faster SSC actions (2). Collectively, these findings highlight the need for further research to clarify the relationship between inertia and jump performance, particularly with individualized inertia prescription. This approach could allow users to perform jump movements at higher velocities and improve transfer to unloaded jumps. Additionally, higher movement velocities (shorter durations) and greater eccentric strength are associated with improved vertical jump performance (119, 120), which can be achieved through flywheel training and may help to optimize performance. However, these hypotheses require further investigation.

It might be expected that our inertial range was sufficient to observe the relationship between inertia and COD. For example, Munoz-Lopez et al. (121) found that moments of inertia from 0.01 to $0.2\text{ kg}\cdot\text{m}^2$ were optimal for achieving eccentric overload in peak power and peak velocity variables, which could theoretically transfer to high-velocity deceleration actions in the COD task. In our study, inertial ranges were mostly within these frames. However, improving COD performance may not require eccentric overload, as traditional resistance training was effective across both low ($\geq 30\text{--}<69\%$ 1-RM, $\text{SMD} = 0.76$) and high ($\geq 70\%$ 1-RM, $\text{SMD} = 0.77$) loads (122). Interpretation should be cautious here, as the study did not separate youth and young adults. These findings suggest that high loads may not be necessary to enhance COD in flywheel training, but further research is needed to clarify the role of inertial load in COD adaptation.

For sprint performance, the relationship between inertia and adaptation may depend on the athlete's responses to load direction and unilateral exercise. Sprint ability relies on unilateral limb involvement and horizontal vector components (2). In our review, we were unable to include unilateral limb involvement in our model due to asymmetric data distribution, nor account for load vector direction because of limited studies. Evidence suggests that loads aligned with sprint-specific vectors (horizontal) may improve transfer to sprinting (2). The specific vector adaptation in the flywheel training remains limited and is only partially addressed at the individual study level. For example, de Hoyo et al. (84) found horizontal flywheel loading improved short-distance sprints (10 m) but not longer sprints (20 m) compared to traditional vertical loading. Moreover, Skok et al. (82) showed that multidirectional vector loading with predominantly horizontal vector exercises produced similar results to vertical vector in flywheel training. However, these are single-flywheel studies, and their generalizability is uncertain. In traditional resistance training, findings are also mixed: a study showed no difference between vertical and horizontal exercises (e.g., half squats, hip thrusts) across sprint distances (123). Another study reported that

horizontal loading improved short sprint distances (≤ 10 m, ES = 0.72, $p = 0.01$) but not longer sprint distances (≥ 20 m, $p = 0.83$) (124). Given these inconsistencies, limb involvement, load direction, and inertia magnitude may influence adaptations across different sprint distances. Future flywheel studies should consider all these factors to identify the most effective combinations for sprint performance.

Overall, statistically significant effects of inertial load were observed only for eccentric power in this meta-analysis. Although the regression coefficients were large, there was substantial variation (wide confidence intervals). These findings may be attributed to the lack of individualization in inertial loading. Many studies used constant inertia as the method for intensity allocation, which could result in different perceived intensity zones for each participant and lead to varied responses to the assigned load and potentially interact with training volume, as volume cannot be considered independently of the assigned load. Consequently, insufficient individualization of inertia may be a significant issue, potentially confounding the findings and adversely affecting dose–response relationships. Again, future studies should individualize inertia for each participant based on the recommended concentric velocity or power values (125) and investigate volume and inertia responses while controlling for other relevant flywheel training moderators within the same model. This approach acknowledges that training adaptations result from the interaction of multiple variables, not just a few isolated factors.

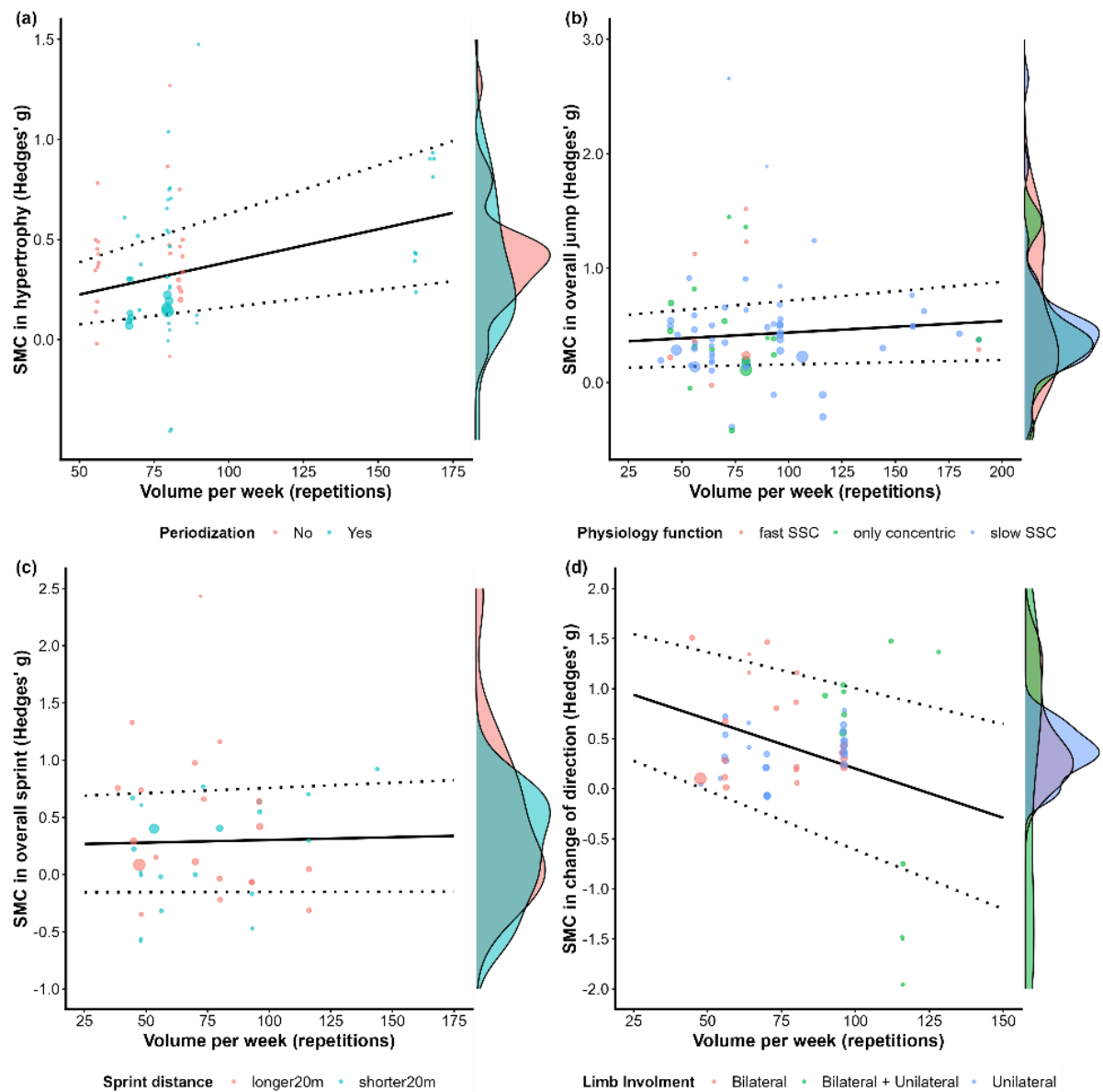


Figure 4. The moderating effects of (a) weekly training volume on hypertrophy, (b) overall jump, (c) overall sprint, and (d) change of direction, visualized from the final multilevel meta-regression models. The plotted relationships are adjusted for all moderators in the model (a) (inertia, limb involvement, periodization, and volume-inertia interaction), (b) (inertia, training level, limb involvement, periodization, physiological function of jump, and volume-inertia interaction), (c) (inertia, periodization, sprint distance, and volume-inertia interaction), and (d) (inertia, limb involvement, periodization, and volume-inertia interaction). Solid lines represent the estimated effect, and dotted lines show the 95% confidence interval. Points are weighed by the inverse of their variance.

4.3. GRADE AND Risk of Bias Assessment

Previous reviews reported wide confidence intervals, indicating imprecision in results, for pre-post effects of flywheel training and comparisons with traditional resistance training across strength, power, hypertrophy, jump, sprint, and COD (24, 25, 86, 87, 126). Similar to our study, Petré et al. reported high heterogeneity (I^2), reflecting low consistency in strength, power, jump, and sprint outcomes (24). No subgroup analyses were performed in those reviews to explore the source heterogeneity, as in other comparable studies (25).

Regarding risk of bias, most studies did not report randomization or allocation concealment, limiting the assessment of selection bias. Although we treated studies as observational, the absence of randomization may allow systematic differences between groups. Approximately half of the studies reported reasons for dropouts and adherence criteria; for the remainder, incomplete data could not be properly judged. Only two studies pre-registered their methodologies to assess selective reporting bias. Effort bias also requires attention: while most studies encouraged maximal concentric effort, not all reported momentum build-up repetitions, which can confound the number of targeted maximal repetitions. Reporting maximal and submaximal repetitions separately would be good practice for flywheel studies. Additionally, at least two familiarization sessions are advised (127), but only ~50% of studies followed this. Flywheel training may also interact with concurrent resistance or endurance programs targeting the same muscle groups, potentially limiting or confounding adaptations. A dose-response gradient was observed in hypertrophy (weekly volume) and eccentric power (inertia), which helps to establish a potential cause-and-effect relationship (128). Overall, the certainty of evidence ranged from very low to moderate, depending on the outcome. Future studies should address these factors to improve methodological qualities and reduce bias.

4.4. Practical, Research, and Training Prescription Recommendations Based on Longitudinal Study Findings

Practical Recommendations

Practitioners aiming to enhance both concentric and eccentric strength and power can effectively use flywheel training. Increasing weekly training volume does not appear to provide additional benefits for concentric or eccentric strength. To further improve concentric strength, bilateral exercises are recommended, and assessments should match the type of limb involvement (bilateral, unilateral, or both). Flywheel training is more effective for highly trained populations in improving concentric power, and flywheel-specific assessments should be used to monitor power changes. Higher inertias improve eccentric power, with the highest inertia in this review being $0.11 \text{ kg}\cdot\text{m}^2$; however, combining high training volumes with high inertia may reduce their effectiveness due to a negative interaction. Flywheel training supports hypertrophy, with greater weekly training volumes likely enhancing adaptations by bringing participants closer to muscular failure. The transferability of flywheel resistance training to the performance of athletic tasks: jump, sprint, and COD seems limited; integrating sport- or movement-specific exercises, such as unloaded jumps and sprints, may enhance these adaptations. Both low and high weekly volumes and inertias can provide sufficient stimulus to improve jump, sprint, and COD performance.

Training Prescription, Reporting, and Future Research Directions

Training programs and future studies should utilize inertial load profiling based on velocity or power values to individualize intensity, recognizing that responses to flywheel training are highly individual. The set and repetition ranges observed in this review were effective for adaptation (Table 3); however, higher training volumes led to greater improvements in hypertrophy. Therefore, prescribing additional sets and intra-set repetitions may further enhance hypertrophy.

Future studies should accurately report the inertia used by each participant. When inertia is prescribed based on velocity or power, the average inertia should be reported rather than ranges. After determining individual inertial loads, the volume load equation ($\text{sets} \times \text{repetitions} \times \text{inertia}$) should be applied to understand interactions with training volume.

Research should further explore dose-response relationships by varying sets, repetitions, weekly frequency (1–3 sessions), inertial loads, and velocity- or power-based prescriptions. Training to failure, periodization strategies, and their effects on strength, power, hypertrophy, jump, sprint, and COD performance should also be investigated. Given the limited transfer to athletic tasks, combining flywheel training with sport-specific movements and appropriate vectorial loading may enhance outcomes.

Finally, eccentric overload ratios should be tracked and reported for each participant to clarify whether eccentric overload contributes to performance. While prior work suggested it may not be necessary for force or hypertrophy adaptations (90), further research is needed to confirm these findings.

4.5. Limitations and Considerations

The current study has several limitations that should be considered when interpreting the findings. First, more accurate reporting of inertial ranges is needed. When velocity and power values are used for profiling, studies often report only the range rather than the exact values, which may not reflect the true average inertia of the training group and could have introduced errors in our inertia and volume-inertia interaction calculations. Second, eccentric overload in flywheel training is technique-dependent (8), and true eccentric overload is not guaranteed unless electric motor-supported flywheels are used (80, 90). When non-motorised flywheels are employed, monitoring session force, power, or velocity is necessary to verify eccentric overload and better understand its adaptation, though practical limitations may prevent such tracking in some settings. Third, we did not specify or use the shaft type as a categorical moderator, which can affect the mechanical outputs (129) and potentially influence results; future studies should explore this when sufficient data are available. Fourth, our findings represent combined male and female data; sex-specific responses should be investigated in future research. Fifth, missing moderator data in some studies limited inclusion in meta-regressions, emphasizing the need for comprehensive reporting of all training variables. Sixth, errors may have occurred during data extraction from figures. Researchers are encouraged to share raw data or respond to data requests, enabling others to calculate correlation coefficients and experimental changes in standard deviation. Seventh, studies should be more precise in reporting repetitions being used to build up momentum, and the maximum repetitions performed (e.g., 6 [+2] repetitions), as variations in this approach may cause minor deviations in volume calculations.

5. Conclusions

Flywheel training provides an effective stimulus for strength and power development. To enhance strength, particularly concentric strength, bilateral exercises should be prioritized over unilateral and mixed-limb approaches, and the testing method should match the limb involvement used in training. Highly trained athletes can benefit more from flywheel training for improving concentric power, which should be evaluated using flywheel-specific assessments to accurately capture adaptations. Higher inertial loads support eccentric power development. However, they should not be combined with higher training volumes, as volume and inertia negatively interact. In contrast, flywheel training has small effects on hypertrophy and the performance of athletic tasks. For hypertrophy, higher sets and intra-set repetitions are recommended, given the observed dose-response relationship. For the performance of athletic tasks, incorporating sport- and movement-specific jump, sprint, and COD drills may help overcome the limited transferability from flywheel training. The absence of a dose-response relationship suggests that lower training volumes may still be effective for the performance of athletic tasks. Inertial load was observed to be effective only for improving eccentric power. The lack of significant effects in other variables may be due to insufficient individualization of inertial load, leading to varied intensity perceptions. Overall, higher training volumes are important for hypertrophy, whereas volume does not appear decisive for strength, power, or athletic task performance.

Supplementary information: Supplementary materials are available online and via the Open Science Framework (OSF, osf.io/s3qgr).

Declarations

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Code availability: Not applicable.

Author contributions: DB performed all analyses, visualized the data (with visualization code provided by IJ for figures 3 and 4), and wrote the first draft of the manuscript. All authors contributed to editing and revising the manuscript and approved the final version.

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